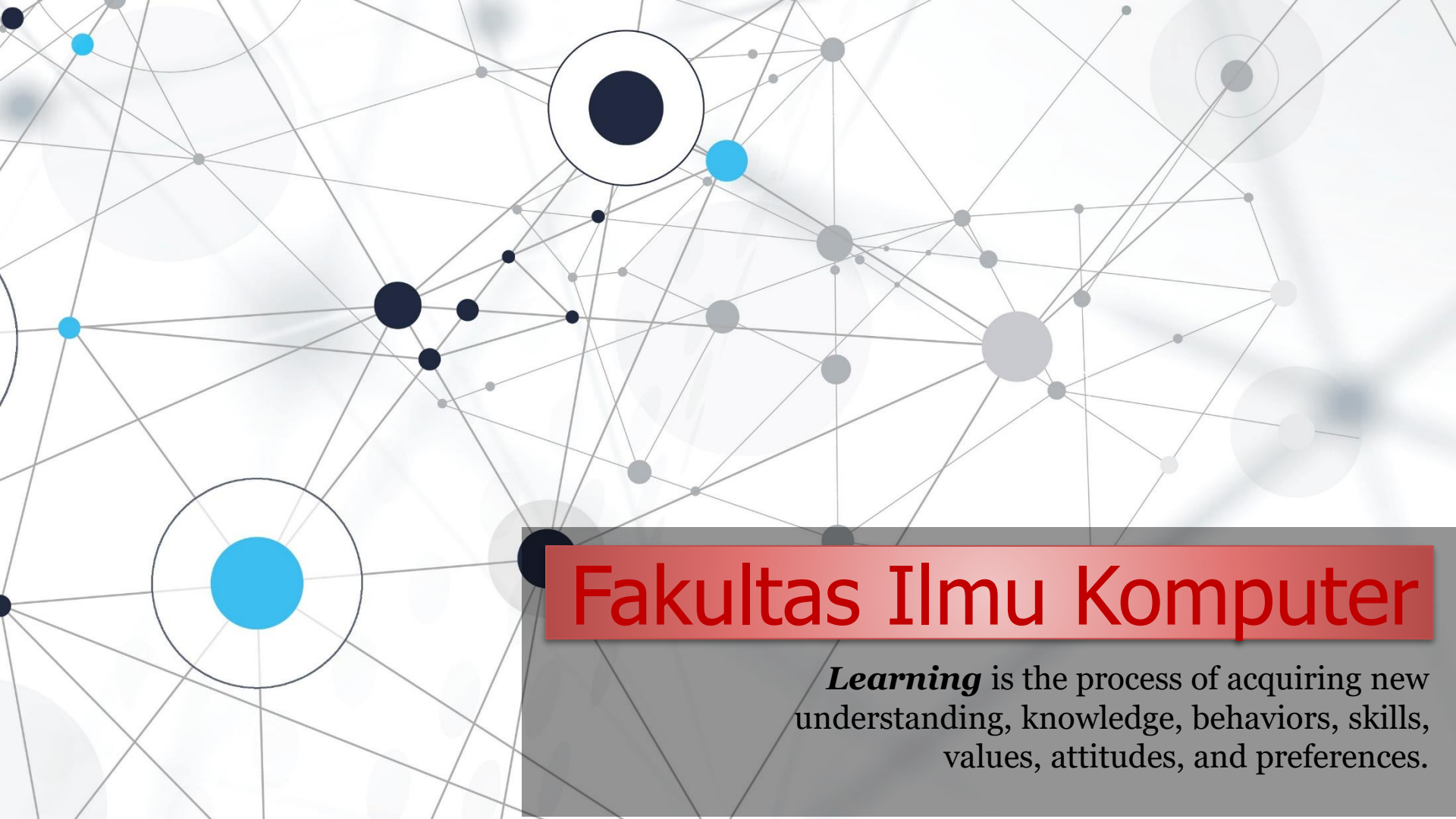


An abstract network diagram with various sized nodes (black, blue, grey) connected by thin grey lines. Some nodes are highlighted with larger circles. The background is white with faint grey circles.

Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

(Class Practical #1)

Monitoring, Logging & Security



Welcome to *the* DevOps Engineering *course*

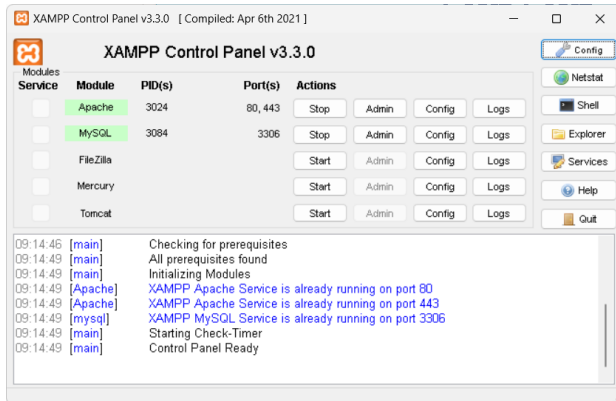
A decorative border of green leaves and branches at the top of the slide.

1 – Create New Database & Tables

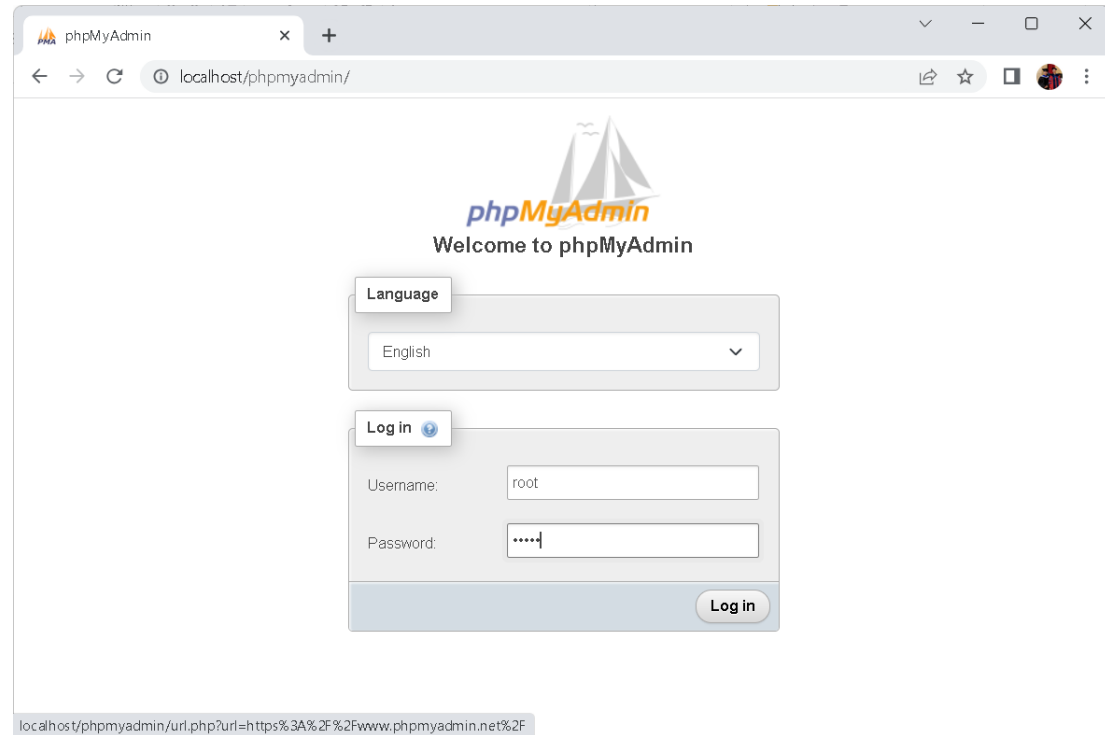
A decorative border of green leaves and branches at the bottom of the slide.

Start Services & Sign In

Xampp Control Panel

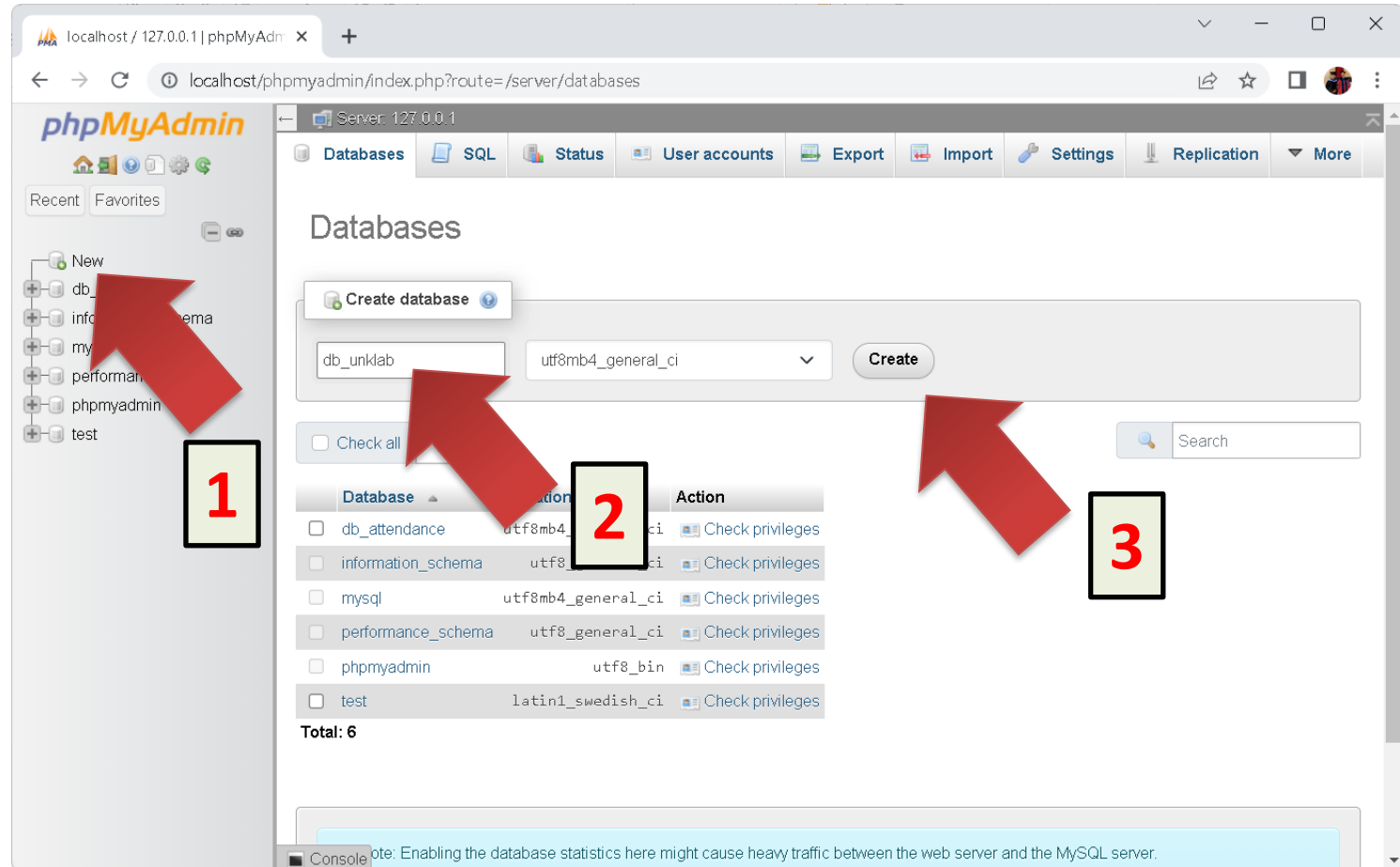


PHPMyAdmin Log In Page



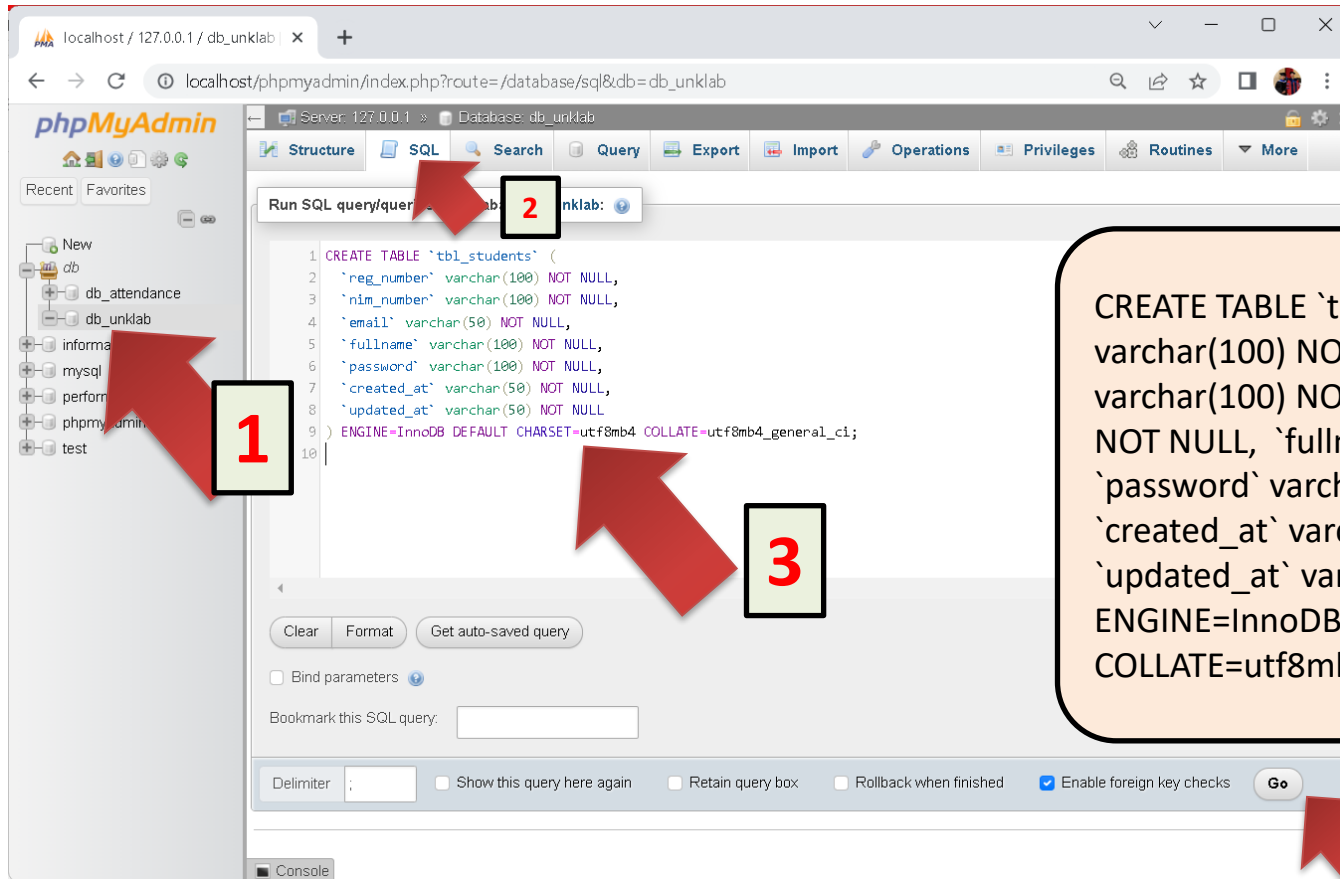
Halaman PHPMyAdmin **Log In** adalah halaman login yang digunakan untuk mengakses *user interface* web **PHPMyAdmin**, yang digunakan untuk mengelola dan mengakses database MySQL melalui *interface web* atau GUI.

Create New Database Name



Buat database baru dengan nama “***db_unklab***”, selanjutnya click button “***create***” untuk melanjutkan proses.

Create Table “tbl_students”



CREATE TABLE `tbl\_students` (`reg\_number` varchar(100) NOT NULL, `nim\_number` varchar(100) NOT NULL, `email` varchar(50) NOT NULL, `fullname` varchar(100) NOT NULL, `password` varchar(100) NOT NULL, `created\_at` varchar(50) NOT NULL, `updated\_at` varchar(50) NOT NULL) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;

Pastikan bahwa kalian sudah click pada database name “*db_unklab*” dan selanjutnya click tab “SQL”.

Create Table “tbl_operator”

localhost / 127.0.0.1 / db_unklab

localhost/phpmyadmin/index.php?route=/database/sql&db=db_unklab

Server: 127.0.0.1 » Database: db_unklab

Structure SQL Search Query Export Import Operations Privileges Routines More

Run SQL query/query

```
1 CREATE TABLE `tbl_operator` (  
2   `nip` int(11) NOT NULL,  
3   `email` varchar(100) NOT NULL,  
4   `fullname` varchar(150) NOT NULL,  
5   `password` varchar(100) NOT NULL,  
6   `phone_number` varchar(50) NOT NULL,  
7   `role` varchar(30) NOT NULL,  
8   `created_at` varchar(100) NOT NULL,  
9   `updated_at` varchar(100) NOT NULL  
10  ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

Clear Format Get auto-saved query

☐ Bind parameters

Bookmark this SQL query:

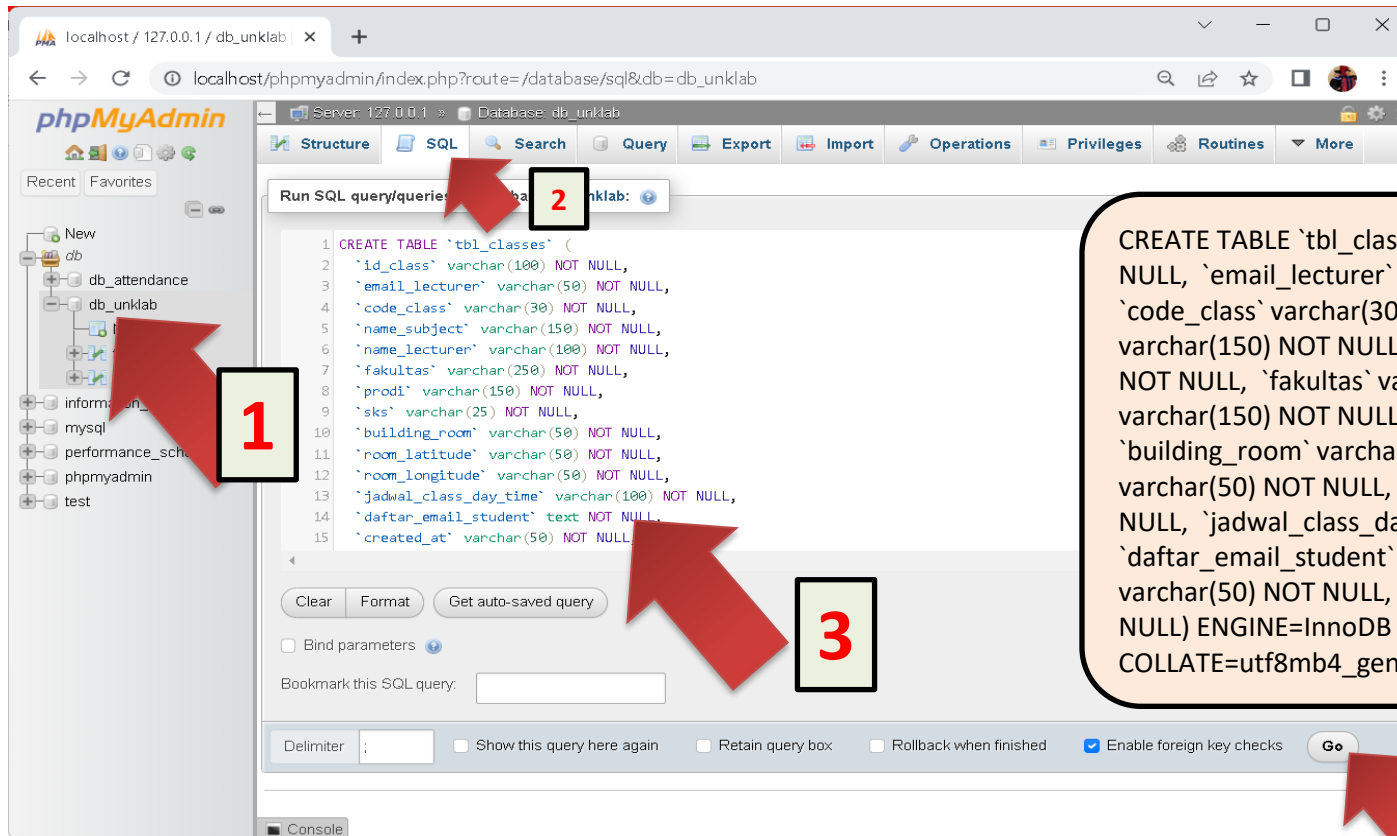
Delimiter: ☐ Show this query here again ☐ Retain query box ☐ Rollback when finished ☒ Enable foreign key checks Go

Console

```
CREATE TABLE `tbl_operator` (  
  `nip` int(11) NOT NULL,  
  `email` varchar(100) NOT NULL,  
  `fullname` varchar(150) NOT NULL,  
  `password` varchar(100) NOT NULL,  
  `phone_number` varchar(50) NOT NULL, `role`  
  varchar(30) NOT NULL, `created_at`  
  varchar(100) NOT NULL, `updated_at`  
  varchar(100) NOT NULL) ENGINE=InnoDB  
  DEFAULT CHARSET=utf8mb4  
  COLLATE=utf8mb4_general_ci;
```

Pastikan bahwa anda harus click pada database name “*db_unklab*” dan selanjutnya click tab “SQL”.

Create Table “tbl_classes”



```
CREATE TABLE `tbl_classes` ( `id_class` varchar(100) NOT NULL, `email_lecturer` varchar(50) NOT NULL, `code_class` varchar(30) NOT NULL, `name_subject` varchar(150) NOT NULL, `name_lecturer` varchar(100) NOT NULL, `fakultas` varchar(250) NOT NULL, `prodi` varchar(150) NOT NULL, `sks` varchar(25) NOT NULL, `building_room` varchar(50) NOT NULL, `room_latitude` varchar(50) NOT NULL, `room_longitude` varchar(50) NOT NULL, `jadwal_class_day_time` varchar(100) NOT NULL, `daftar_email_student` text NOT NULL, `created_at` varchar(50) NOT NULL, `status_class` varchar(30) NOT NULL) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

Pastikan bahwa anda harus click pada database name “db_unklab” dan selanjutnya click tab “SQL”.

Create Table “tbl_attendance_list”

The screenshot shows the phpMyAdmin interface for a database named 'db_unklab'. The left sidebar shows a tree of databases, with 'db_unklab' selected (indicated by a red arrow and a box labeled '1'). The main area shows the 'SQL' tab selected (indicated by a red arrow and a box labeled '2'). The SQL query editor contains the following code:

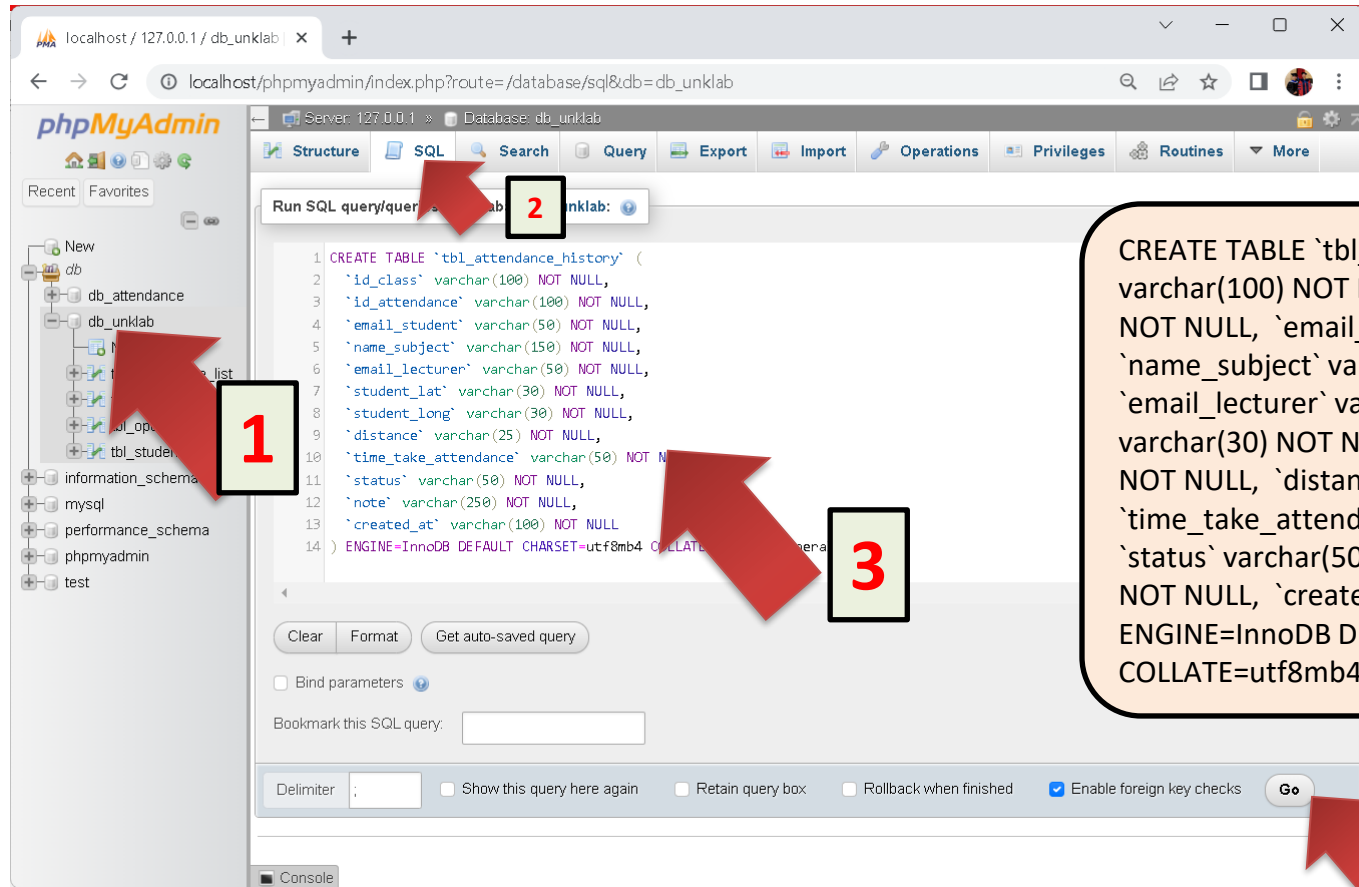
```
1 CREATE TABLE `tbl_attendance_list` (  
2   `id_attendance` varchar(30) NOT NULL,  
3   `title_short` varchar(100) NOT NULL,  
4   `date_attendance` varchar(50) NOT NULL,  
5   `time_attendance` varchar(50) NOT NULL,  
6   `id_class` varchar(30) NOT NULL,  
7   `name_subject` varchar(100) NOT NULL,  
8   `email_lecturer` varchar(70) NOT NULL,  
9   `name_lecturer` varchar(70) NOT NULL,  
10  `room_latitude` varchar(50) NOT NULL,  
11  `room_longitude` varchar(50) NOT NULL,  
12  `max_radius` varchar(50) NOT NULL,  
13  `created_at` varchar(30) NOT NULL  
14 ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

Below the query editor, there are buttons for 'Clear', 'Format', and 'Get auto-saved query'. There is also a 'Bookmark this SQL query' field. At the bottom, there are checkboxes for 'Show this query here again', 'Retain query box', 'Rollback when finished', and 'Enable foreign key checks' (which is checked). A 'Go' button is at the bottom right (indicated by a red arrow and a box labeled '4').

```
CREATE TABLE `tbl_attendance_list` (  
  `id_attendance` varchar(30) NOT NULL, `title_short` varchar(100) NOT  
  NULL, `date_attendance` varchar(50) NOT NULL,  
  `time_attendance` varchar(50) NOT NULL, `id_class`  
  varchar(30) NOT NULL, `name_subject` varchar(100)  
  NOT NULL, `email_lecturer` varchar(70) NOT NULL,  
  `name_lecturer` varchar(70) NOT NULL,  
  `room_latitude` varchar(50) NOT NULL,  
  `room_longitude` varchar(50) NOT NULL,  
  `max_radius` varchar(50) NOT NULL, `created_at`  
  varchar(30) NOT NULL) ENGINE=InnoDB DEFAULT  
  CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

Pastikan bahwa anda harus click pada database name “*db_unklab*” dan selanjutnya click tab “SQL”.

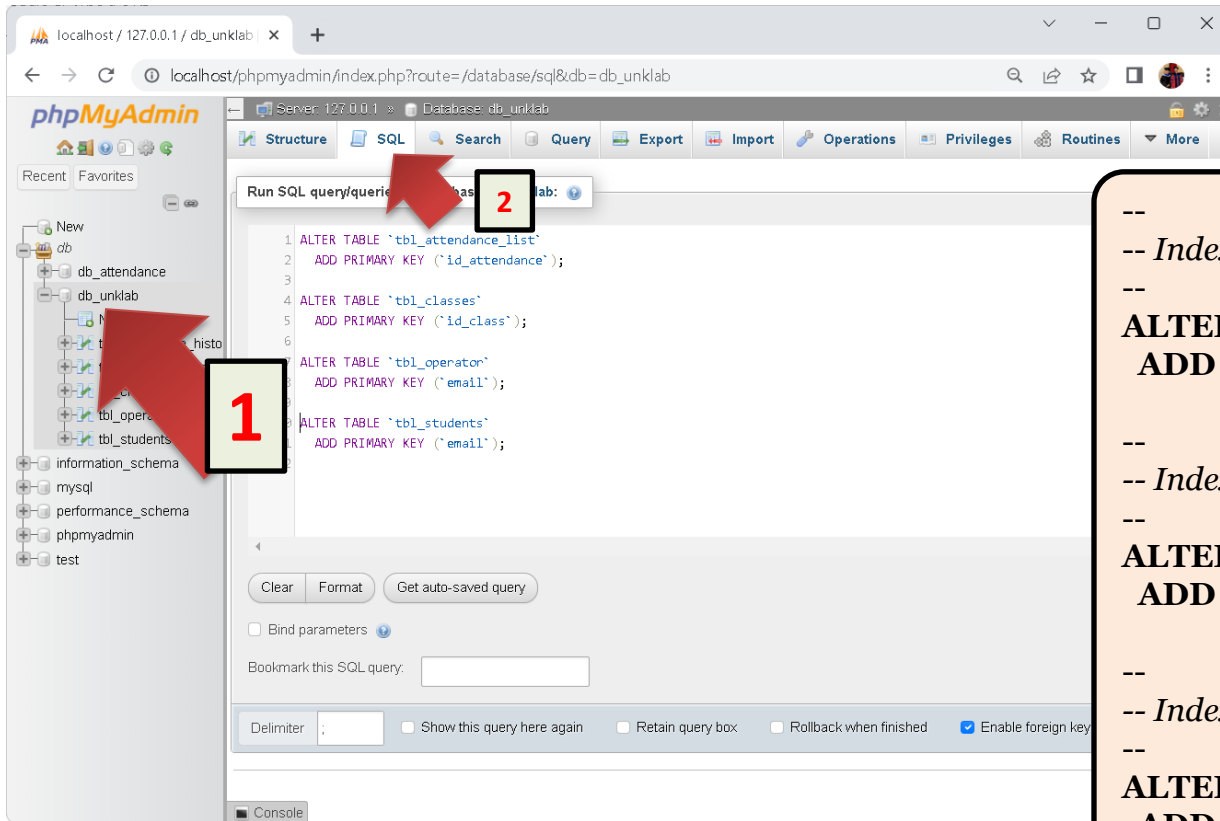
Create Table “tbl_attendance_history”



CREATE TABLE `tbl\_attendance\_history` (
 `id\_class` varchar(100) NOT NULL, `id\_attendance` varchar(100) NOT NULL, `email\_student` varchar(50) NOT NULL, `name\_subject` varchar(150) NOT NULL, `email\_lecturer` varchar(50) NOT NULL, `student\_lat` varchar(30) NOT NULL, `student\_long` varchar(30) NOT NULL, `distance` varchar(25) NOT NULL, `time\_take\_attendance` varchar(50) NOT NULL, `status` varchar(50) NOT NULL, `note` varchar(250) NOT NULL, `created\_at` varchar(100) NOT NULL)
ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;

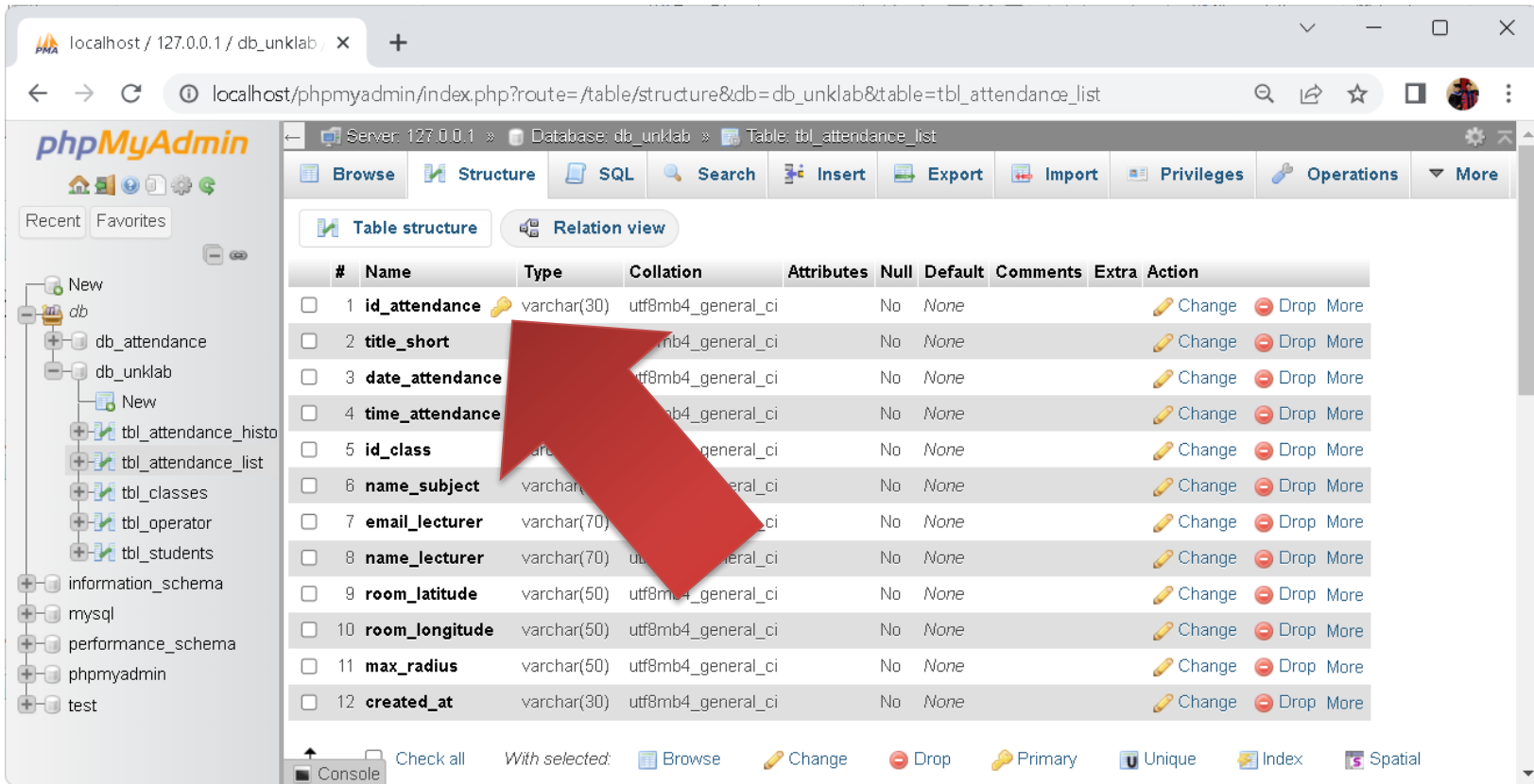
Pastikan bahwa anda harus *click* pada database name “db_unklab” dan selanjutnya *click* tab “SQL”.

Add Primary Key in Tables



```
--  
-- Indexes for table `tbl_attendance_list`  
--  
ALTER TABLE `tbl_attendance_list`  
  ADD PRIMARY KEY (`id_attendance`);  
  
--  
-- Indexes for table `tbl_classes`  
--  
ALTER TABLE `tbl_classes`  
  ADD PRIMARY KEY (`id_class`);  
  
--  
-- Indexes for table `tbl_operator`  
--  
ALTER TABLE `tbl_operator`  
  ADD PRIMARY KEY (`email`);  
  
--  
-- Indexes for table `tbl_students`  
--  
ALTER TABLE `tbl_students`  
  ADD PRIMARY KEY (`email`);
```

Check Primary Keys in Tables



The screenshot shows the phpMyAdmin interface for a database named 'db_unklab'. The table 'tbl_attendance_list' is selected, and its structure is displayed. The table has 12 columns. The first column, 'id_attendance', is marked as the primary key with a key icon. A large red arrow points to this icon. The table structure is as follows:

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	id_attendance	varchar(30)	utf8mb4_general_ci		No	None			Change Drop More
2	title_short	varchar(30)	utf8mb4_general_ci		No	None			Change Drop More
3	date_attendance	date	utf8mb4_general_ci		No	None			Change Drop More
4	time_attendance	time	utf8mb4_general_ci		No	None			Change Drop More
5	id_class	varchar(30)	utf8mb4_general_ci		No	None			Change Drop More
6	name_subject	varchar(30)	utf8mb4_general_ci		No	None			Change Drop More
7	email_lecturer	varchar(70)	utf8mb4_general_ci		No	None			Change Drop More
8	name_lecturer	varchar(70)	utf8mb4_general_ci		No	None			Change Drop More
9	room_latitude	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
10	room_longitude	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
11	max_radius	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
12	created_at	varchar(30)	utf8mb4_general_ci		No	None			Change Drop More

At the bottom of the interface, there are buttons for 'Check all', 'With selected', 'Browse', 'Change', 'Drop', 'Primary', 'Unique', 'Index', and 'Spatial'.

Structure Tables *in* Database

The screenshot shows the phpMyAdmin interface for the database `db_unklab`. The left sidebar displays the database structure, including `db_unklab` and its tables: `tbl_attendance_history`, `tbl_attendance_list`, `tbl_classes`, `tbl_operator`, and `tbl_students`. The main panel shows the 'Structure' tab for the database. A table of 5 tables is displayed, each with 16.0 K B size and InnoDB engine. The table names and their corresponding actions are listed below.

Table	Action	Rows	Type	Collation	Size	Over
☐ <code>tbl_attendance_history</code>	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 K B	
☐ <code>tbl_attendance_list</code>	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 K B	
☐ <code>tbl_classes</code>	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 K B	
☐ <code>tbl_operator</code>	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 K B	
☐ <code>tbl_students</code>	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 K B	
5 tables	Sum	0	InnoDB	utf8mb4_general_ci	80.0 K B	

Below the table, there is a 'Check all' checkbox and a 'With selected:' dropdown menu. At the bottom, there is a 'Create new table' button and a form to create a new table with fields for 'Table name' and 'Number of columns' (set to 4).

Terdapat 5 *tables* yang telah dibuat pada `db_unklab`.

The slide features a white background with green leaves and branches framing the top and bottom edges. The leaves are vibrant green with some brownish edges, suggesting they might be from a tree like a Japanese Quince.

2 – Insert Data *in* Tables

How to Insert Data in Table?

The screenshot shows the phpMyAdmin interface for a database named 'db_unklab'. The left sidebar shows the database structure, with 'tbl_students' selected under 'db_unklab'. The main area displays the 'SQL' tab, where an SQL query is entered. The query is an INSERT statement for the 'tbl_students' table, with values for 'reg_number', 'nim_number', 'email', 'fullname', 'password', 'created_at', and 'updated_at'. The query is highlighted with a red box labeled '2'. The 'Go' button at the bottom right is highlighted with a red box labeled '4'. The 'SELECT' tab is highlighted with a red box labeled '1'. The 'tbl_students' table name in the sidebar is highlighted with a red box labeled '3'.

1

2

3

4

Untuk menginputkan data ke dalam *table*, pastikan bahwa anda harus click pada ***table name*** yang menjadi target dan selanjutnya click tab “SQL”.

Insert Table “tbl_student”

Data for table `tbl\_students`

```
INSERT INTO `tbl_students` (`reg_number`, `nim_number`, `email`, `fullname`, `password`, `created_at`, `updated_at`) VALUES('123456789012', '154321', 's154321@student.unklab.ac.id', 'Radix Basalama', '12345', '2023-04-13 22:54:33', '2023-04-13 22:54:33'),('105021710031', '21710268', 's21710268@student.unklab.ac.id', 'Pongantung, Jeremy', '1234qwer', '12-12-2023 10:32-43', '12-12-2023 10:32-43'),('123456789012', '254321', 's254321@student.unklab.ac.id', 'Putra Podomi', '12345', '2023-04-13 22:54:33', '2023-04-13 22:54:33'),('123456789012', '354321', 's354321@student.unklab.ac.id', 'ChinChin Long', '12345', '2023-04-13 22:54:33', '2023-04-13 22:54:33'),('123456789012', '454321', 's454321@student.unklab.ac.id', 'Miehie Kawekiang Sie', '12345', '2023-04-13 22:54:33', '2023-04-13 22:54:33'),('123456789012', '554321', 's554321@student.unklab.ac.id', 'Dinda Manopo Putri', '12345', '2023-04-13 22:54:33', '2023-04-13 22:54:33'),('123456789012', '604321', 's604321@student.unklab.ac.id', 'Margono Raharjo', '12345', '2023-04-13 22:54:33', '2023-04-13 22:54:33'),('123456789012', '644321', 's644321@student.unklab.ac.id', 'Sempoyonangan Ojomiso', '12345', '2023-04-13 22:54:33', '2023-04-13 22:54:33'),('123456789012', '654321', 's654321@student.unklab.ac.id', 'David Kawekiang', '12345', '12-12-2023 10:32-43', '12-12-2023 10:32-43'),('123456789012', '754321', 's754321@student.unklab.ac.id', 'Citra Lentari Anada', '12345', '2023-04-13 22:54:33', '2023-04-13 22:54:33'),('123456789012', '854321', 's854321@student.unklab.ac.id', 'Sandra Dewi', '12345', '2023-04-13 22:54:33', '2023-04-13 22:54:33'),('123456789012', '954321', 's954321@student.unklab.ac.id', 'Putri Moningka', '12345', '2023-04-13 22:54:33', '2023-04-13 22:54:33');
```

Insert Table “tbl_operator”

Data for table `tbl\_operator`

```
INSERT INTO `tbl_operator` (`nip`, `email`, `fullname`, `password`, `phone_number`, `role`, `created_at`, `updated_at`) VALUES(12345, 'andrewws@unklab.ac.id', 'Andrew Liem', 'abcd123', '0852123456', 'admin', '12-04-2023 12:04:32', '12-04-2023 12:04:32'),(12345, 'gledis@unklab.ac.id', 'Gladis Sepang', 'abcw1243', '0852123456', 'admin', '12-04-2023 12:04:32', '12-04-2023 12:04:32'),(876578, 'juanmambu@unklab.ac.id', 'Juan Taju', 'acbc1213', '0852123456', 'admin', '12-04-2023 12:04:32\r\n', '12-04-2023 12:04:32\r\n'),(34567, 'reymond@unklab.ac.id', 'Reymond Rotikan', '123475', '0852123456', 'staff', '12-04-2023 12:04:32', '12-04-2023 12:04:32'),(12345, 'semmytaju@unklab.ac.id', 'Semmy Wellem Taju', 'abc1233', '0852123456', 'admin', '12-04-2023 12:04:32', '12-04-2023 12:04:32'),(23456, 'stenlyadam@unklab.ac.id', 'Stenly Adam', '123425', '0852123456', 'admin', '12-04-2023 12:04:32', '12-04-2023 12:04:32');
```

Insert Table “tbl_classes”

Data for table `tbl\_classes`

```
INSERT INTO `tbl_classes` (`id_class`, `email_lecturer`, `code_class`, `name_subject`, `name_lecturer`, `fakultas`, `prodi`, `sks`, `building_room`,  
`room_latitude`, `room_longitude`, `jadwal_class_day_time`, `daftar_email_student`, `created_at`, `status_class`) VALUES('CLASS64381629449e2',  
'semmy@unklab.ac.id', '234ABCD', 'Java Programming', 'Semmy Wellem Taju', 'Fakultas Ekonomi', 'Pendidikan Keagamaan Kristen (Religious)', '3 SKS', 'GA Computer  
Lab 202/FIK', '1,4178155', '124,9844905', 'Selasa ~ Kamis, 08.00 - 10.00',  
'semmy@unklab.ac.id,yonatan@gmail.com,jeremy@yahoo.com,s254321@student.unklab.ac.id,s354321@student.unklab.ac.id,s454321@student.unklab.ac.id,tajusemmy@  
yahoo.com,budiman@yahoo.com', '2023-04-13 22:48:47', 'complete'),('CLASS64381721a4505', 'semmy@unklab.ac.id', '2345ABC', 'Python Programming', 'Adam Stelly',  
'Fakultas Pendidikan', 'Pendidikan Luar Sekolah (Non Formal Education)', '3 SKS', 'GK1-204 Comp. Lab (Hardware)', '1,4178155', '124,9844905', 'Senin ~ Rabu, 08.00 -  
09.00', 's254321@student.unklab.ac.id,juan@yahoo.com,putra@gmail.com', '2023-04-13 22:54:33', 'complete'),('CLASS64383858ec58d', 'semmy@unklab.ac.id',  
'AI3456S', 'Artificial Intelligence', 'Green Sandag', 'Fakultas Ilmu Komputer', 'Sistem Informasi (Information System)', '3 SKS', 'GK3-205', '1,4178155', '124,9844905', 'Senin  
~ Rabu, 14.30 - 16.00',  
'bobon@yhaoo.com,jeremy@yahoo.com,s154321@student.unklab.ac.id,s254321@student.unklab.ac.id,s354321@student.unklab.ac.id,semmy@yahoo.com,s454321@stude  
nt.unklab.ac.id,budihardo@ggamil.com,s554321@student.unklab.ac.id,patricia@yahoo.com,s644321@student.unklab.ac.id,s654321@student.unklab.ac.id,s754321@stude  
nt.unklab.ac.id', '2023-04-14 01:14:59', 'active'),('CLASS643838975a476', 'semmy@unklab.ac.id', 'PD23232S', 'Statistics and Probability', 'Semmy Wellem Taju', 'Fakultas  
Ilmu Komputer', 'Sistem Informasi (Information System)', '3 SKS', 'GK2-Tissue Culture Lab', '1,4178155', '124,9844905', 'Senin ~ Rabu, 08.00 - 09.00',  
's154321@student.unklab.ac.id,moningkatri@gmail.com,brendaabu@yahoo.com,saturundes@gmail.com,eugenie@gmail.com,semmy@yahoo.com,s754321@student.unkla  
b.ac.id,s954321@student.unklab.ac.id,ayu@yahoo.com,mandola@gamil.com,budi@gmail.chot.com', '2023-04-14 01:15:47', 'active'),('CLASS643a940726a3d',  
'semmy@unklab.ac.id', 'MLC4335', 'Machine Learning Algorithms', 'Semmy Wellem Taju', 'Fakultas Ilmu Komputer', 'Pendidikan Keagamaan Kristen (Religious)', '3 SKS',  
'GA Computer Lab 203/FIK', '1,4178155', '124,9844905', 'Selasa ~ Kamis, 07.00 - 08.30', 'semmy@yahoo.com,tajudespies@gmail.com', '2023-04-15 20:10:28',  
'active'),('CLASS643aa53d7f8c9', 'semmy@unklab.ac.id', 'DBA984', 'Big Data Analysis', 'Stelly Adam', 'Fakultas Ilmu Komputer', 'Sistem Informasi (Information System)',  
'3 SKS', 'GK3-203', '1,4191753', '124,9845743', 'Senin ~ Rabu, 17.00 - 18.00',  
'semmy@unklab.ac.id,s604321@student.unklab.ac.id,s654321@student.unklab.ac.id,s854321@student.unklab.ac.id,arnold@gmail.com,mawar@student.unklab.ac.id,sem  
mytaju@yahoo.com', '2023-04-15 21:23:54', 'active'),('CLASS643ac4d56e4a8', 'semmy@unklab.ac.id', 'FED3464', 'Front-end Web Development', 'Raissa Maringka',  
'Akademi Sekretari Manajemen Indonesia', 'Pendidikan Keagamaan Kristen (Religious)', '3 SKS', 'GA Computer Lab 202/FIK', '1,4174176', '124,9840658', 'Senin ~ Rabu,  
14.30 - 16.00', 'budy@gmail.com,s21710268@student.unklab.ac.id,jeremy@unklab.ac.id,semmy@yahoo.com', '2023-04-15 23:38:52', 'active'),('CLASS6445e32a2c21e',  
'juan@unklab.ac.id', 'CN1264D', 'Computer Networking', 'Juan Taju', 'Fakultas Ilmu Komputer', 'Sistem Informasi (Information System)', '3 SKS', 'GA Computer Lab  
202/FIK', '1,4174176', '124,9840658', 'Senin ~ Rabu, 10.00 - 12.00', ' ', '2023-04-24 10:03:03', 'active'),('CLASS6445e35cd31eb', 'juan@unklab.ac.id', 'HCI5644', 'Human  
Computer Interaction', 'Juan Taju', 'Fakultas Keperawatan', 'Pendidikan Ekonomi (Economy Education)', '3 SKS', 'GK1 Nursing Lab (Demo Room)', '1,4180611',  
'124,9845825', 'Senin ~ Rabu, 09.00 - 10.00', ' ', '2023-04-24 10:03:47', 'complete'),('CLASS6445e3981e4aa', 'juan@unklab.ac.id', 'NLP3456', 'Natural Language  
Processing', 'Juan Taju', 'Fakultas Ilmu Komputer', 'Pendidikan Keagamaan Kristen (Religious)', '3 SKS', 'GK2-104', '1,4171919', '124,9828838', 'Senin ~ Rabu, 09.30 -  
11.00', ' ', '2023-04-24 10:05:11', 'active'),('CLASS6445f99431e6', 'juan@unklab.ac.id', 'ABC564D', 'Data Analytics', 'Juan Taju', 'Fakultas Filsafat', 'Bahasa Inggris  
(English)', '3 SKS', 'GA Computer Lab 202/FIK', '1,4174176', '124,9840658', 'Jumat, 09.00 - 10.00', ' ', '2023-04-24 11:17:17', 'active'),('CLASS6445f87da49d6',  
'semmy@unklab.ac.id', 'GAC3678', 'Sentiment Analysis', 'Semmy Wellem Taju', 'Fakultas Ilmu Komputer', 'Filsafat (Philosophy)', '3 SKS', 'GK2-103', '1,4172063',  
'124,9828819', 'Selasa ~ Kamis, 09.00 - 10.00', 'buddyhars@gmail.com,eugenie@yahoo.com,patricia@gmail.com', '2023-04-24 11:34:05', 'active'),('CLASS6445f8b7ce4de',  
'semmy@unklab.ac.id', 'APC496', 'Fundamental Apache PHP5', 'Semmy Wellem Taju', 'Fakultas Ilmu Komputer', 'Pendidikan Keagamaan Kristen (Religious)', '3 SKS',  
'GK2-103', '1,4172063', '124,9828819', 'Senin ~ Rabu, 09.30 - 11.00', 'semmy@yahoo.com,putra@gmail.com,tampijeni@yahoo.com', '2023-04-24 11:34:57', 'active');
```

Insert Table “tbl_attendance_list”

Data for table `tbl\_attendance\_list`

```
INSERT INTO `tbl_attendance_list` (`id_attendance`, `title_short`, `date_attendance`, `time_attendance`, `id_class`, `name_subject`, `email_lecturer`,  
`name_lecturer`, `room_latitude`, `room_longitude`, `max_radius`, `created_at`) VALUES('ATD665ac4d56e4b7', 'Attendance #1', '21 April 2023', '10.30 - 11.30',  
'CLASS663ac4d56e4a8', 'Research Method', 'andrew@unklab.ac.id', 'Andrew Liem', '1,4174176', '124,9840658', '20', '12-12-2023 10:32:43'), ('ATD745ac4d56e4b7',  
'Attendance #1', '17 April 2023', '14.00 - 15.00', 'CLASS643ac4d56e4a8', 'Big Data Analysis', 'semmy@unklab.ac.id', 'Semmy Wellem Taju', '1,4174176', '124,9840658', '100',  
'12-12-2023 10:32:43'), ('ATD745ac4d56e4c7', 'Attendance #2', '26 April 2023', '09.30 - 10.40', 'CLASS643ac4d56e4a8', 'System Analysis and Design',  
'semmy@unklab.ac.id', 'Semmy Wellem Taju', '1,20438484', '124,343566', '200', '12-12-2023 10:32:43'), ('ATD788ac4d56e4b7', 'Attendance #1', '20 April 2023', '08.30 -  
19.30', 'CLASS883ac4d56e4a8', 'Front-end Web Development', 'reymond@unklab.ac.id', 'Reymond Rotikan', '1,4174176', '124,9840658', '50', '12-12-2023 10:32-  
43'), ('ATD995ac4d56e4b7', 'Attendance #1', '19 April 2023', '07.30 - 08.30', 'CLASS993ac4d56e4a8', 'Computer Networkings', 'stelly@unklab.ac.id', 'Stelly Adam',  
'1,4174176', '124,9840658', '40', '12-12-2023 10:32:43'), ('ATTE6444526a2e1b7', 'Attendance #3', '26 April 2023', '08.00 - 10.00', 'CLASS643aa53d7f8c9', 'Big Data  
Analysis', 'semmy@unklab.ac.id', 'Stelly Adam', '1,4191753', '124,9845743', '200', '2023-04-23 05:32:50'), ('ATTE64445305d6cee', 'Attendance #3', '29 April 2023', '07.00 -  
08.30', 'CLASS643ac4d56e4a8', 'Front-end Web Development', 'semmy@unklab.ac.id', 'Raissa Marangka', '1,4174176', '124,9840658', '250', '2023-04-23  
05:35:45'), ('ATTE6445b2210cf5c', 'Attendance #15', '18 April 2023', '20.00 - 21.00', 'CLASS643aa53d7f8c9', 'Big Data Analysis', 'semmy@unklab.ac.id', 'Stelly Adam',  
'1,4191753', '124,9845743', '150', '2023-04-24 06:33:58'), ('ATTE6445f3cb39bca', 'Attendance #1', '24 April 2023', '07.00 - 08.30', 'CLASS6445e3981e4aa', 'Natural Language  
Processing', 'juan@unklab.ac.id', 'Juan Taju', '1,4171919', '124,9828838', '100', '2023-04-24 11:13:42'), ('ATTE64462606d35a3', 'Attendance #1', '24 April 2023', '08.00 -  
09.00', 'CLASS64383858ec58d', 'Artificial Intelligence', 'semmy@unklab.ac.id', 'Green Sandag', '1,4178155', '124,9844905', '50', '2023-04-24  
14:47:56'), ('ATTE6446261c86db6', 'Attendance #2', '26 April 2023', '10.00 - 11.00', 'CLASS64383858ec58d', 'Artificial Intelligence', 'semmy@unklab.ac.id', 'Green Sandag',  
'1,4178155', '124,9844905', '60', '2023-04-24 14:48:44'), ('ATTE644647fba4842', 'Attendance #4', '30 April 2023', '13.00 - 14.30', 'CLASS643ac4d56e4a8', 'Front-end Web  
Development', 'semmy@unklab.ac.id', 'Raissa Marangka', '1,4174176', '124,9840658', '70', '2023-04-24 17:12:55'), ('ATTE6447363ab836f', 'Attendance #5', '26 April 2023',  
'08.00 - 09.00', 'CLASS643ac4d56e4a8', 'Front-end Web Development', 'semmy@unklab.ac.id', 'Raissa Marangka', '1,4174176', '124,9840658', '70', '2023-04-25  
10:11:08'), ('ATTE644754987cf94', 'Attendance #1', '25 April 2023', '07.00 - 08.30', 'CLASS6445f8b7ce4de', 'Fundamental Apache PHP5', 'semmy@unklab.ac.id', 'Semmy  
Wellem Taju', '1,4172063', '124,9828819', '60', '2023-04-25 12:18:56'), ('ATTE644754b0cdcab', 'Attendance #2', '27 April 2023', '08.00 - 09.00', 'CLASS6445f8b7ce4de',  
'Fundamental Apache PHP5', 'semmy@unklab.ac.id', 'Semmy Wellem Taju', '1,4172063', '124,9828819', '150', '2023-04-25 12:19:13'), ('ATTE6451d686525a0', 'Attendance  
#1', '04 May 2023', '07.00 - 08.30', 'CLASS643838975a476', 'Statistics and Probability', 'semmy@unklab.ac.id', 'Semmy Wellem Taju', '1,4178155', '124,9844905', '60',  
'2023-05-03 11:36:12'), ('ATTE6451d6b52ba51', 'Attendance #2', '04 May 2023', '08.00 - 09.00', 'CLASS643838975a476', 'Statistics and Probability', 'semmy@unklab.ac.id',  
'Semmy Wellem Taju', '1,4178155', '124,9844905', '70', '2023-05-03 11:36:46'), ('ATTE6451d6ce98d25', 'Pertemuan ke-3', '06 May 2023', '08.00 - 09.00',  
'CLASS643838975a476', 'Statistics and Probability', 'semmy@unklab.ac.id', 'Semmy Wellem Taju', '1,4178155', '124,9844905', '60', '2023-05-03  
11:37:06'), ('ATTE6451daee51boa', 'Attendance #3', '04 May 2023', '08.00 - 09.00', 'CLASS6445f8b7ce4de', 'Fundamental Apache PHP5', 'semmy@unklab.ac.id', 'Semmy  
Wellem Taju', '1,4172063', '124,9828819', '70', '2023-05-03 11:54:45'), ('ATTE6451db0537079', 'Attendance #4', '05 May 2023', '07.00 - 09.00', 'CLASS6445f8b7ce4de',  
'Fundamental Apache PHP5', 'semmy@unklab.ac.id', 'Semmy Wellem Taju', '1,4172063', '124,9828819', '60', '2023-05-03 11:54:57'), ('ATTE6451db115ba2c', 'Attendance  
#5', '04 May 2023', '07.00 - 09.00', 'CLASS6445f8b7ce4de', 'Fundamental Apache PHP5', 'semmy@unklab.ac.id', 'Semmy Wellem Taju', '1,4172063', '124,9828819', '60',  
'2023-05-03 11:55:08'), ('ATTE6451db1ca165d', 'Attendance #6', '08 May 2023', '08.00 - 09.00', 'CLASS6445f8b7ce4de', 'Fundamental Apache PHP5',  
'semmy@unklab.ac.id', 'Semmy Wellem Taju', '1,4172063', '124,9828819', '80', '2023-05-03 11:55:23'), ('ATTE6451db64f1047', 'Pertemuan ke-4', '05 May 2023', '07.00 -  
09.00', 'CLASS643838975a476', 'Statistics and Probability', 'semmy@unklab.ac.id', 'Semmy Wellem Taju', '1,4178155', '124,9844905', '60', '2023-05-03  
11:56:36'), ('ATTE6451db74d7759', 'Pertemuan ke-5', '05 May 2023', '07.00 - 09.00', 'CLASS643838975a476', 'Statistics and Probability', 'semmy@unklab.ac.id', 'Semmy  
Wellem Taju', '1,4178155', '124,9844905', '70', '2023-05-03 11:56:46');
```

Insert Table “tbl_attendance_history”

Data for table `tbl\_attendance\_history`

```
INSERT INTO `tbl_attendance_history` (`id_class`, `id_attendance`, `email_student`, `name_subject`, `email_lecturer`, `student_lat`, `student_long`, `distance`, `time_take_attendance`, `status`, `note`, `created_at`) VALUES('CLASS643ac4d56e4a8', 'ATD745ac4d56e4b7', 's354321@student.unklab.ac.id', 'Artificial Intelligence', 'semmy@unklab.ac.id', '1,239484', '124,348583', '10', '10:15:49', 'P', '', '2023-04-13 22:54:33'),('CLASS643ac4d56e4a8', 'ATD745ac4d56e4b7', 's154321@student.unklab.ac.id', 'Artificial Intelligence', 'semmy@unklab.ac.id', '1,239484', '124,348583', '16', '10:00:49', 'P', '', '2023-04-13 22:54:33'),('CLASS643ac4d56e4a8', 'ATD745ac4d56e4b7', 's254321@student.unklab.ac.id', 'Artificial Intelligence', 'semmy@unklab.ac.id', '1,239484', '124,348583', '12', '10:05:49', 'P', '', '2023-04-13 22:54:33'),('CLASS643ac4d56e4a8', 'ATD745ac4d56e4b7', 's354321@student.unklab.ac.id', 'Artificial Intelligence', 'semmy@unklab.ac.id', '1,239484', '124,348583', '20', '10:08:49', 'P', '', '2023-04-13 22:54:33'),('CLASS643ac4d56e4a8', 'ATD745ac4d56e4b7', 's454321@student.unklab.ac.id', 'Artificial Intelligence', 'semmy@unklab.ac.id', '1,239484', '124,348583', '15', '10:20:49', 'P', '', '2023-04-13 22:54:33'),('CLASS643ac4d56e4a8', 'ATD745ac4d56e4b7', 's554321@student.unklab.ac.id', 'Artificial Intelligence', 'semmy@unklab.ac.id', '1,239484', '124,348583', '6', '10:30:49', 'P', '', '2023-04-13 22:54:33'),('CLASS643ac4d56e4a8', 'ATD745ac4d56e4b7', 's644321@student.unklab.ac.id', 'Artificial Intelligence', 'semmy@unklab.ac.id', '1,239484', '124,348583', '7', '10:45:49', 'P', '', '2023-04-13 22:54:33'),('CLASS643ac4d56e4a8', 'ATD745ac4d56e4b7', 's654321@student.unklab.ac.id', 'Artificial Intelligence', 'semmy@unklab.ac.id', '1,239484', '124,348583', '8', '10:27:49', 'P', '', '2023-04-13 22:54:33'),('CLASS643838975a476', 'ATTE6451d6ce98d25', 's154321@student.unklab.ac.id', 'Statistics and Probability', 'semmy@unklab.ac.id', 'o', 'o', 'o', 'o', 'P', 'Hadir', '2023-05-03 11:37:39'),('CLASS643838975a476', 'ATTE6451d6ce98d25', 's754321@student.unklab.ac.id', 'Statistics and Probability', 'semmy@unklab.ac.id', 'o', 'o', 'o', 'o', 'L', 'Datang terlambat', '2023-05-03 11:37:58'),('CLASS643838975a476', 'ATTE6451d6ce98d25', 'mandola@gmail.com', 'Statistics and Probability', 'semmy@unklab.ac.id', 'o', 'o', 'o', 'o', 'E', 'Sakit', '2023-05-03 11:38:20'),('CLASS6445f8b7ce4de', 'ATTE6451db1ca165d', 'semmy@yahoo.com', 'Fundamental Apache PHP5', 'semmy@unklab.ac.id', 'o', 'o', 'o', 'o', 'P', '', '2023-05-03 11:55:46'),('CLASS6445f8b7ce4de', 'ATTE6451db1ca165d', 'putra@gmail.com', 'Fundamental Apache PHP5', 'semmy@unklab.ac.id', 'o', 'o', 'o', 'o', 'E', 'Sakit', '2023-05-03 11:55:54');
```




3 – Backup Database [MySQL PHPMyAdmin]

How to Backup Database MySQL?

- 1) **Buka PHPMyAdmin:** Akses halaman PHPMyAdmin melalui browser dengan memasukkan URL yang sesuai. Biasanya, alamat *default* adalah ***http://localhost/phpmyadmin*** atau sesuai dengan konfigurasi server.
- 2) **Masuk ke PHPMyAdmin:** Masukkan nama *users* (*username*) dan kata sandi (*password*) MySQL yang valid untuk masuk ke antarmuka PHPMyAdmin.
- 3) **Pilih Database:** Setelah masuk, **pilih database yang ingin di backup** dari panel kiri. Jika tidak ada *database* yang terlihat, pastikan telah membuat dan mengimpor database sebelumnya.
- 4) **Pilih Tab "Export":** Di bagian atas *dashboard* PHPMyAdmin, temukan tab **"Export"** dan klik. Ini secara otomatis akan diarahkan ke halaman ekspor database.

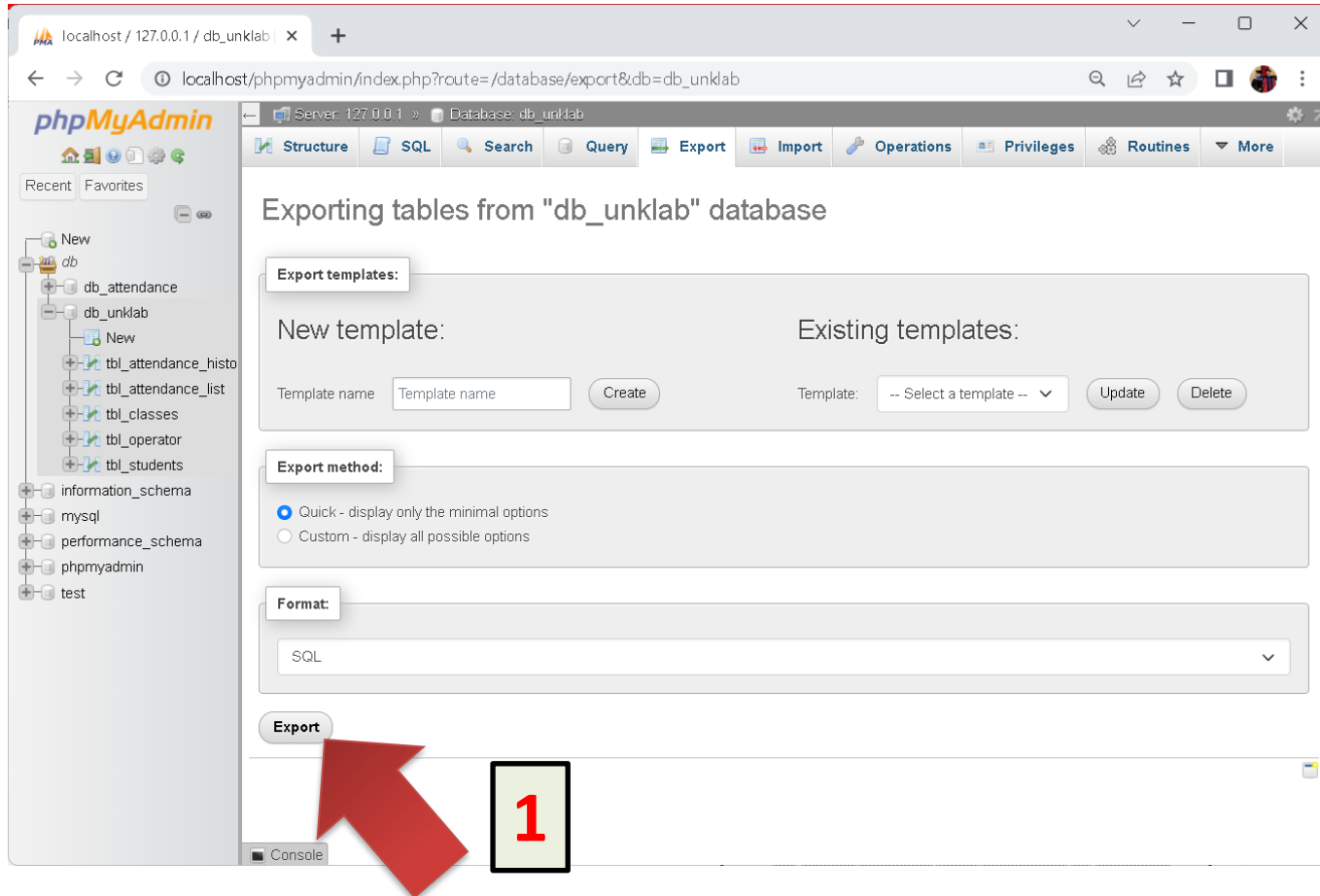
Export MySQL Database

The screenshot shows the phpMyAdmin interface for a MySQL database named 'db_unklab'. The 'Export' tab is selected in the top navigation bar. The left sidebar shows the database structure, with 'db_unklab' selected. The main area displays a table list for 'db_unklab' with 5 tables: 'attendance_history', 'attendance_list', 'ses', 'tbl_operator', and 'tbl_students'. The 'tbl_students' table is selected. The 'Export' tab is highlighted with a red arrow and a '2' in a box. The 'db_unklab' database is highlighted with a red arrow and a '1' in a box.

Table	Action	Rows	Type	Collation	Size	Over
attendance_history	Browse Structure Search Insert Empty Drop	13	InnoDB	utf8mb4_general_ci	16.0 KiB	
attendance_list	Browse Structure Search Insert Empty Drop	24	InnoDB	utf8mb4_general_ci	16.0 KiB	
ses	Browse Structure Search Insert Empty Drop	13	InnoDB	utf8mb4_general_ci	16.0 KiB	
tbl_operator	Browse Structure Search Insert Empty Drop	6	InnoDB	utf8mb4_general_ci	16.0 KiB	
tbl_students	Browse Structure Search Insert Empty Drop	12	InnoDB	utf8mb4_general_ci	16.0 KiB	
5 tables	Sum	68	InnoDB	utf8mb4_general_ci	80.0 KiB	

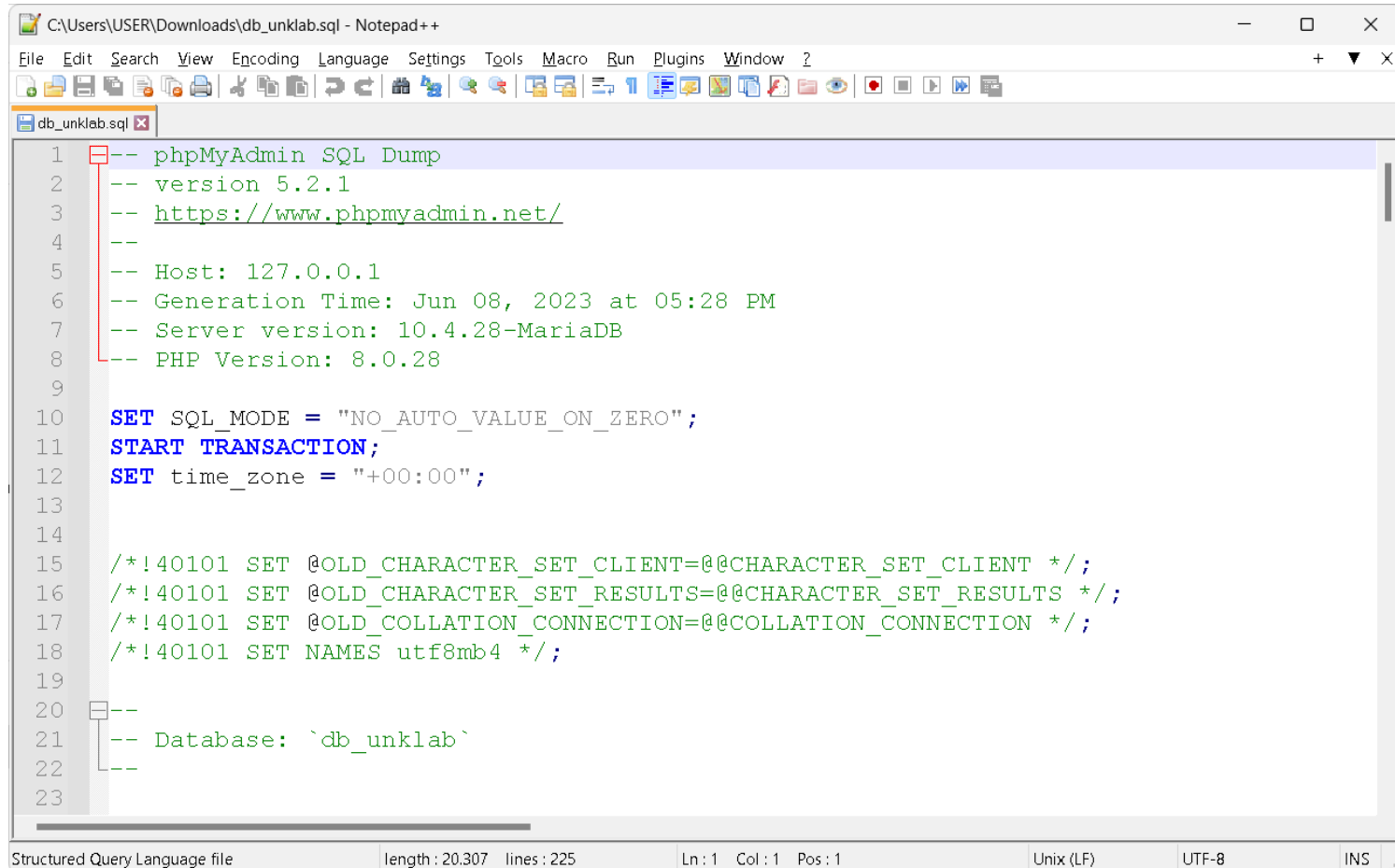
- 1) Pilih Database:** Pertama yang harus dilakukan adalah memilih database yang ingin di backup dari panel kiri. Jika tidak ada database yang terlihat, pastikan telah membuat dan mengimpor database sebelumnya.
- 2) Pilih Tab "Export":** Di bagian atas antarmuka PHPMyAdmin, temukan tab "**Export**" dan klik. Ini secara otomatis akan diarahkan ke halaman ekspor database.

Export MySQL Database



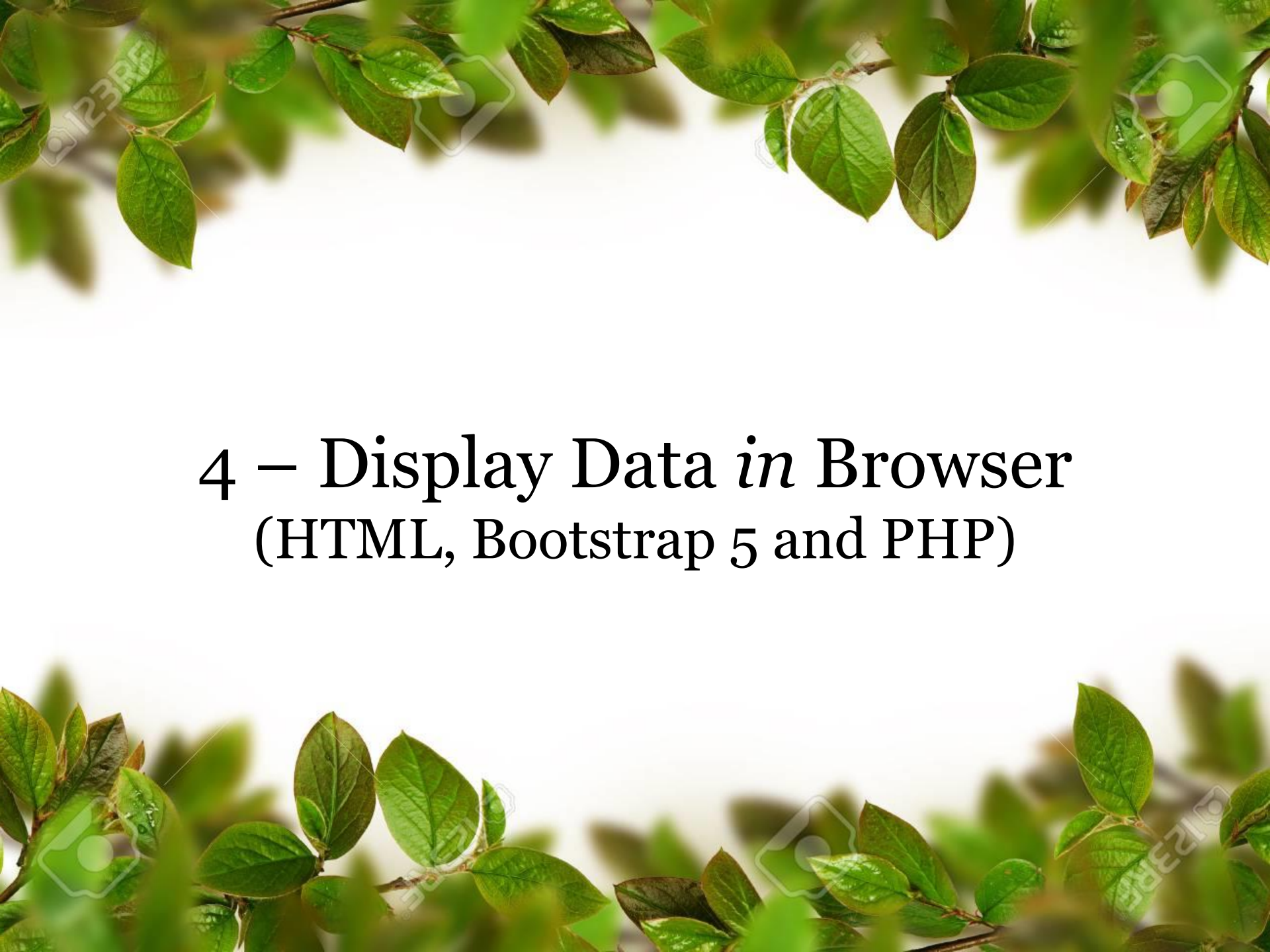
Pilih format *file* yang akan di export, selanjutnya Kalian bisa secara langsung click button “**Export**” dan sistem akan mendownload secara otomatis file dengan nama “***db_unklab.sql***”.

Backup File Downloaded



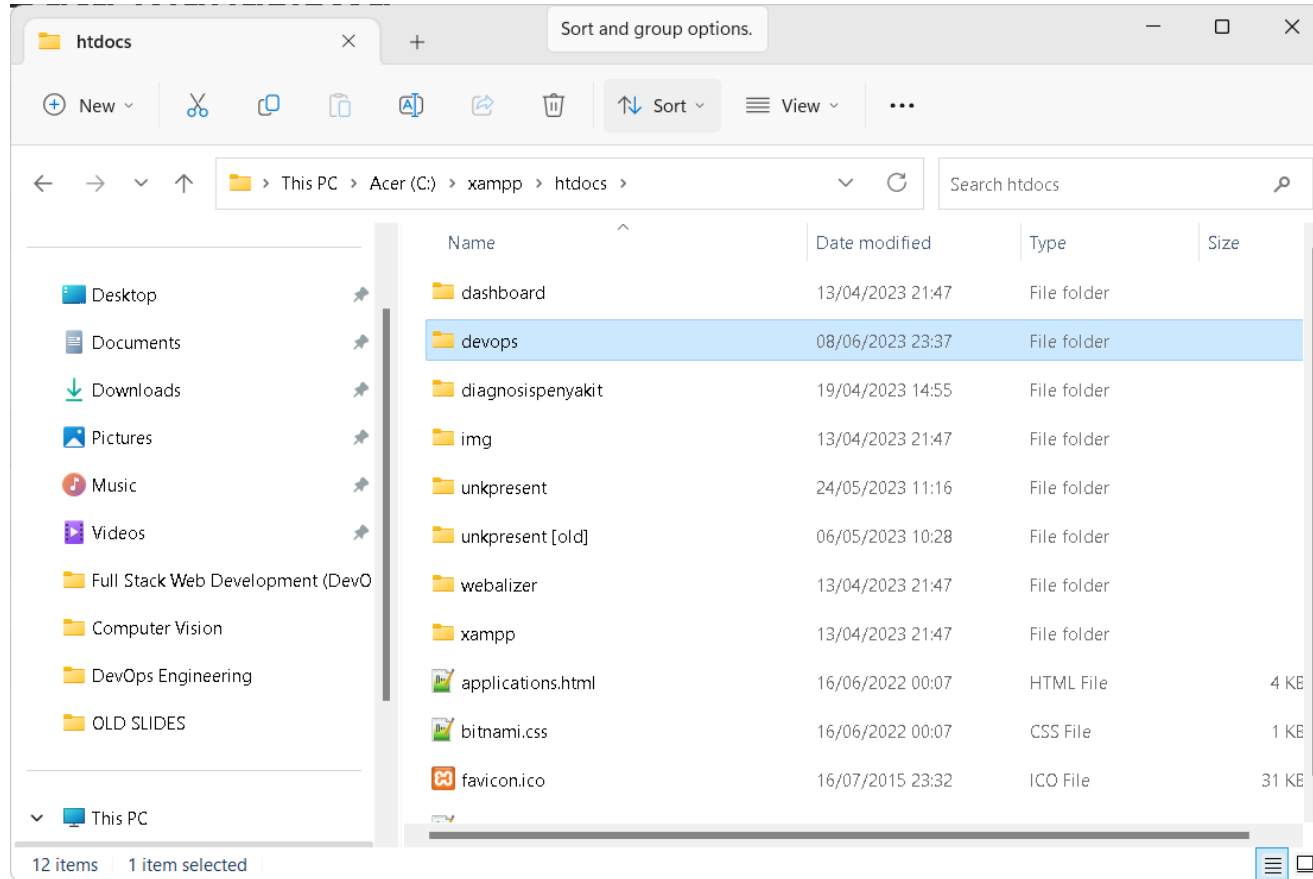
```
C:\Users\USER\Downloads\db_unklab.sql - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
db_unklab.sql
1  -- phpMyAdmin SQL Dump
2  -- version 5.2.1
3  -- https://www.phpmyadmin.net/
4  --
5  -- Host: 127.0.0.1
6  -- Generation Time: Jun 08, 2023 at 05:28 PM
7  -- Server version: 10.4.28-MariaDB
8  -- PHP Version: 8.0.28
9
10 SET SQL_MODE = "NO_AUTO_VALUE_ON_ZERO";
11 START TRANSACTION;
12 SET time_zone = "+00:00";
13
14
15 /*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */;
16 /*!40101 SET @OLD_CHARACTER_SET_RESULTS=@@CHARACTER_SET_RESULTS */;
17 /*!40101 SET @OLD_COLLATION_CONNECTION=@@COLLATION_CONNECTION */;
18 /*!40101 SET NAMES utf8mb4 */;
19
20 --
21 -- Database: `db_unklab`
22 --
23
```

Structured Query Language file length: 20,307 lines: 225 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS



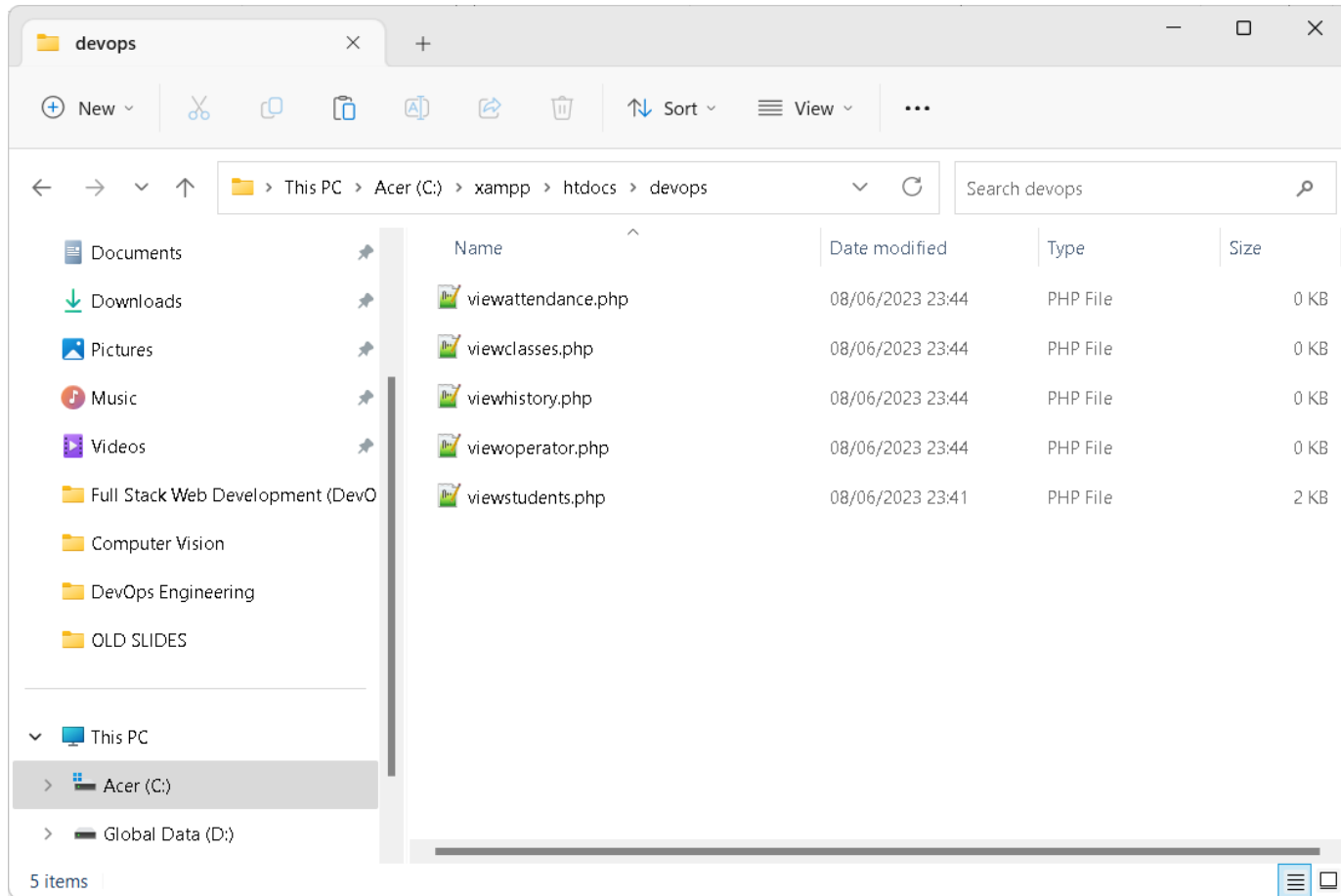
4 – Display Data *in* Browser (HTML, Bootstrap 5 and PHP)

Create New Working Dir



Buat sebuah directory baru “**devops**” di dalam “**htdocs**”.

Create New Files



Buat 5 files di dalam working directory untuk menampilkan data pada setiap table.

Write File “viewstudents.php”

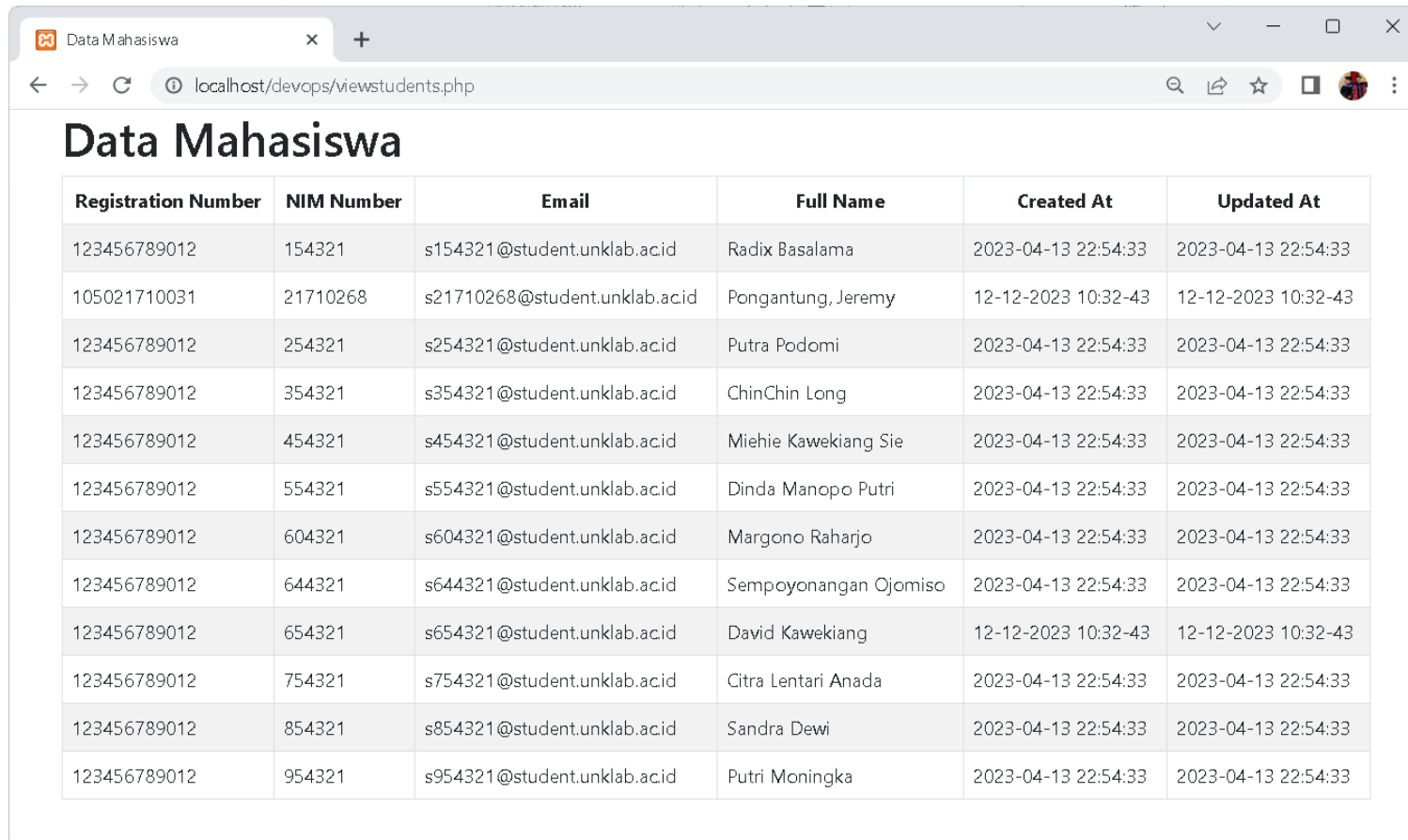
```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Data Mahasiswa</title>
5   <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css">
6 </head>
7 <body>
8   <div class="container">
9     <h1>Data Mahasiswa</h1>
10    <table class="table table-striped table-bordered table-hover">
11      <thead>
12        <tr>
13          <th class="text-center">Registration Number</th>
14          <th class="text-center">NIM Number</th>
15          <th class="text-center">Email</th>
16          <th class="text-center">Full Name</th>
17          <th class="text-center">Created At</th>
18          <th class="text-center">Updated At</th>
19        </tr>
20      </thead>
21      <tbody>
22        <?php
23          // Koneksi ke database
24          $conn = mysqli_connect('localhost', 'root', '12345', 'db_unklab');
25
26          // Periksa koneksi
27          if (!$conn) {
28            die("Koneksi gagal: " . mysqli_connect_error());
29          }
30
31          // Query untuk mendapatkan data mahasiswa
32          $sql = "SELECT * FROM tbl_students";
33          $result = mysqli_query($conn, $sql);
34
35          // Tampilkan data dalam tabel
36          if (mysqli_num_rows($result) > 0) {
37            while ($row = mysqli_fetch_assoc($result)) {
38              echo "<tr>";
39              echo "<td>" . $row['reg_number'] . "</td>";
40              echo "<td>" . $row['nim_number'] . "</td>";
41              echo "<td>" . $row['email'] . "</td>";
42              echo "<td>" . $row['fullname'] . "</td>";
43              echo "<td>" . $row['created_at'] . "</td>";
44              echo "<td>" . $row['updated_at'] . "</td>";
45              echo "</tr>";
```

```
45         echo "</tr>";
46       }
47     } else {
48       echo "<tr><td colspan='6'>Tidak ada data mahasiswa.</td></tr>";
49     }
50
51     // Tutup koneksi
52     mysqli_close($conn);
53   <?>
54   </tbody>
55 </table>
56 </div>
57 </body>
58 </html>
```

Display Data *via* Browser

Open this link in your browser:

<http://localhost/devops/viewstudents.php>



The screenshot shows a web browser window with the title 'Data Mahasiswa'. The address bar displays 'localhost/devops/viewstudents.php'. The main content area features a table with the following data:

Registration Number	NIM Number	Email	Full Name	Created At	Updated At
123456789012	154321	s154321@student.unklab.ac.id	Radix Basalama	2023-04-13 22:54:33	2023-04-13 22:54:33
105021710031	21710268	s21710268@student.unklab.ac.id	Pongantung, Jeremy	12-12-2023 10:32-43	12-12-2023 10:32-43
123456789012	254321	s254321@student.unklab.ac.id	Putra Podomi	2023-04-13 22:54:33	2023-04-13 22:54:33
123456789012	354321	s354321@student.unklab.ac.id	ChinChin Long	2023-04-13 22:54:33	2023-04-13 22:54:33
123456789012	454321	s454321@student.unklab.ac.id	Miehie Kawekiang Sie	2023-04-13 22:54:33	2023-04-13 22:54:33
123456789012	554321	s554321@student.unklab.ac.id	Dinda Manopo Putri	2023-04-13 22:54:33	2023-04-13 22:54:33
123456789012	604321	s604321@student.unklab.ac.id	Margono Raharjo	2023-04-13 22:54:33	2023-04-13 22:54:33
123456789012	644321	s644321@student.unklab.ac.id	Sempoyonangan Ojomiso	2023-04-13 22:54:33	2023-04-13 22:54:33
123456789012	654321	s654321@student.unklab.ac.id	David Kawekiang	12-12-2023 10:32-43	12-12-2023 10:32-43
123456789012	754321	s754321@student.unklab.ac.id	Citra Lentari Anada	2023-04-13 22:54:33	2023-04-13 22:54:33
123456789012	854321	s854321@student.unklab.ac.id	Sandra Dewi	2023-04-13 22:54:33	2023-04-13 22:54:33
123456789012	954321	s954321@student.unklab.ac.id	Putri Moningka	2023-04-13 22:54:33	2023-04-13 22:54:33

"Next, it's your turn!!!"

Modifikasi *code* yang sudah dibuat dan tampilkan semua data yang ada pada *table*:

1) tbl_attendance_history

2) tbl_attendance_list

3) tbl_classes

4) tbl_operator

~~5) tbl_students~~

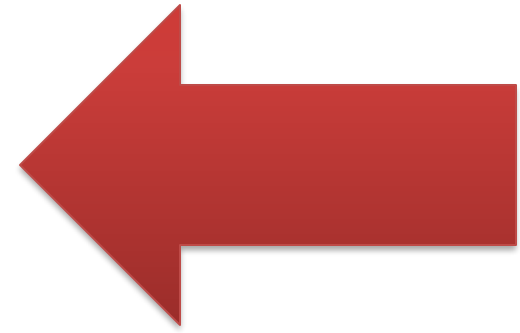
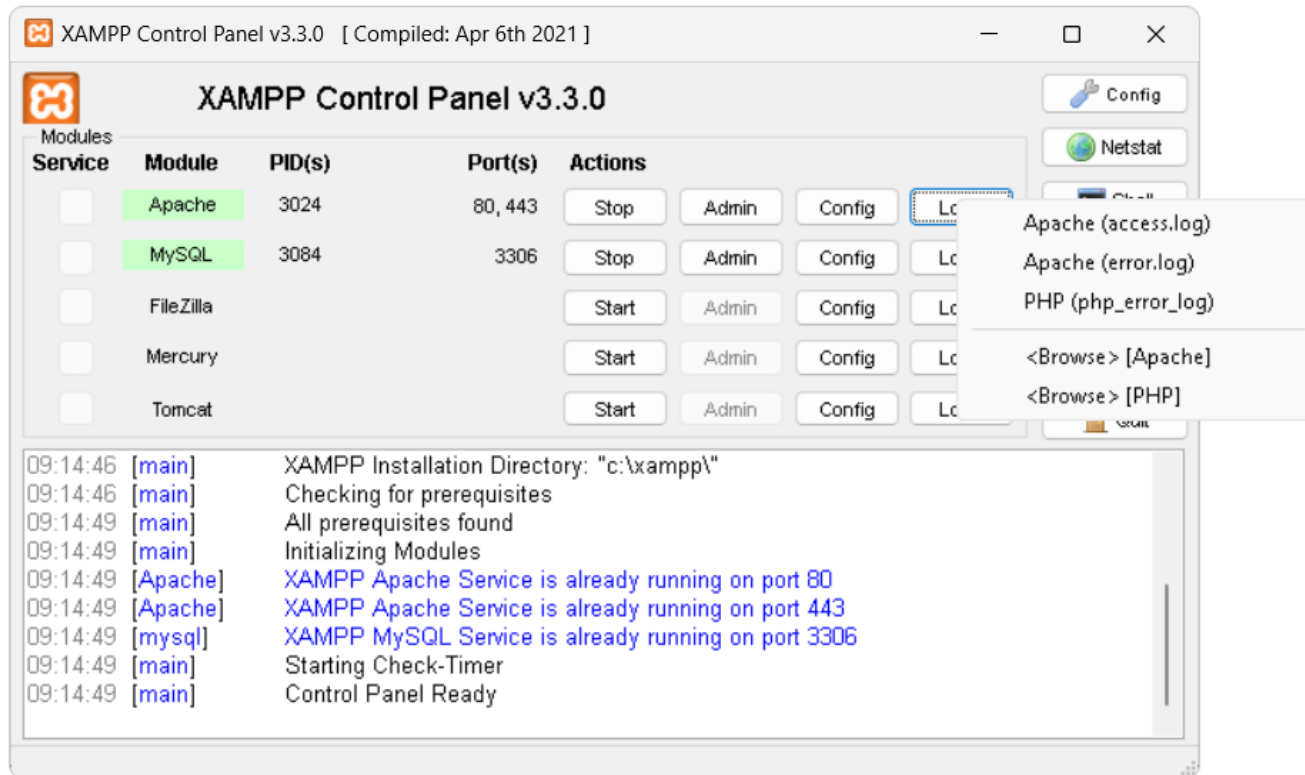
Challenge *for* Students

1. Tambahkan sebuah button “**Insert New Data**” pada masing-masing halaman website yang telah dibuat.
2. Buat penanganan **event handling button** tersebut menggunakan JQuery dan Action Submit Form.
3. Buat penanganan komunikasi ke *server* (*send/request*) menggunakan AJAX dan PHP code.
4. Gunakan type request “POST/GET” atau sesuaikan dengan data yang akan di kirim ke sisi *server* atau diterima dari sisi *server*.

The slide features a white background with green leaves and branches framing the top and bottom edges. The leaves are vibrant green and appear to be from a tree or shrub. The text is centered in the middle of the slide.

5 – Monitoring & Security

Manually View Apache Error Log

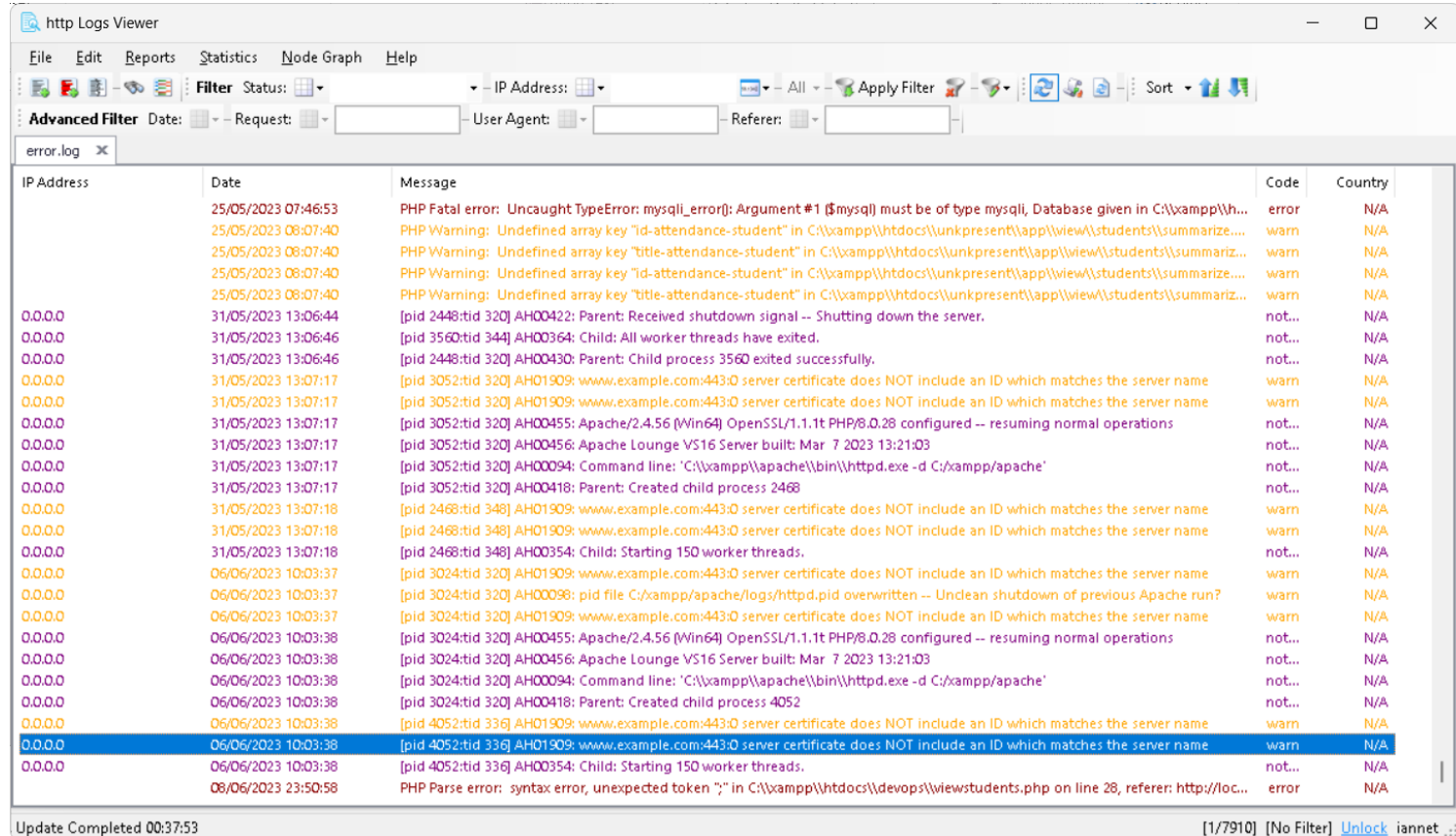


Dari *xampp control panel*, students dapat secara langsung menemukan dan menampilkan *access.log* dan *error.log*

[illegible]

Http Log Viewer adalah *tool* atau *software* yang dirancang khusus untuk membantu dalam membaca, memantau, dan menganalisis **log HTTP server**. Ini memungkinkan kita untuk melihat catatan *log* yang dihasilkan oleh server web, seperti Apache atau Nginx, yang mencatat permintaan dan respons HTTP.

Http Logs Viewer



The screenshot shows the 'http Logs Viewer' application window. It has a menu bar with 'File', 'Edit', 'Reports', 'Statistics', 'Node Graph', and 'Help'. Below the menu is a toolbar with various icons. A 'Filter' section includes 'Status' (a dropdown menu), 'IP Address' (a text input), and 'Apply Filter' button. An 'Advanced Filter' section includes 'Date', 'Request', 'User Agent', and 'Referer' fields. The main area displays a table of log entries. The table has five columns: 'IP Address', 'Date', 'Message', 'Code', and 'Country'. The log entries are color-coded: orange for warnings, purple for informational messages, and red for errors. The status bar at the bottom shows 'Update Completed 00:37:53' and '[1/7910] [No Filter] [Unlock](#) iannet ...'.

IP Address	Date	Message	Code	Country
	25/05/2023 07:46:53	PHP Fatal error: Uncaught TypeError: mysqli_error(): Argument #1 (\$mysql) must be of type mysqli, Database given in C:\xampp\H...	error	N/A
	25/05/2023 08:07:40	PHP Warning: Undefined array key "id-attendance-student" in C:\xampp\htdocs\unkpresent\app\view\students\summarize....	warn	N/A
	25/05/2023 08:07:40	PHP Warning: Undefined array key "title-attendance-student" in C:\xampp\htdocs\unkpresent\app\view\students\summariz...	warn	N/A
	25/05/2023 08:07:40	PHP Warning: Undefined array key "id-attendance-student" in C:\xampp\htdocs\unkpresent\app\view\students\summarize....	warn	N/A
	25/05/2023 08:07:40	PHP Warning: Undefined array key "title-attendance-student" in C:\xampp\htdocs\unkpresent\app\view\students\summariz...	warn	N/A
0.0.0.0	31/05/2023 13:06:44	[pid 2448:tid 320] AH00422: Parent: Received shutdown signal -- Shutting down the server.	not...	N/A
0.0.0.0	31/05/2023 13:06:46	[pid 3560:tid 344] AH00364: Child: All worker threads have exited.	not...	N/A
0.0.0.0	31/05/2023 13:06:46	[pid 2448:tid 320] AH00430: Parent: Child process 3560 exited successfully.	not...	N/A
0.0.0.0	31/05/2023 13:07:17	[pid 3052:tid 320] AH01909: www.example.com:443:O server certificate does NOT include an ID which matches the server name	warn	N/A
0.0.0.0	31/05/2023 13:07:17	[pid 3052:tid 320] AH01909: www.example.com:443:O server certificate does NOT include an ID which matches the server name	warn	N/A
0.0.0.0	31/05/2023 13:07:17	[pid 3052:tid 320] AH00455: Apache/2.4.56 (Win64) OpenSSL/1.1.1t PHP/8.0.28 configured -- resuming normal operations	not...	N/A
0.0.0.0	31/05/2023 13:07:17	[pid 3052:tid 320] AH00456: Apache Lounge VS16 Server built: Mar 7 2023 13:21:03	not...	N/A
0.0.0.0	31/05/2023 13:07:17	[pid 3052:tid 320] AH00094: Command line: 'C:\xampp\apache\bin\httpd.exe -d C:\xampp\apache'	not...	N/A
0.0.0.0	31/05/2023 13:07:17	[pid 3052:tid 320] AH00418: Parent: Created child process 2468	not...	N/A
0.0.0.0	31/05/2023 13:07:18	[pid 2468:tid 348] AH01909: www.example.com:443:O server certificate does NOT include an ID which matches the server name	warn	N/A
0.0.0.0	31/05/2023 13:07:18	[pid 2468:tid 348] AH01909: www.example.com:443:O server certificate does NOT include an ID which matches the server name	warn	N/A
0.0.0.0	31/05/2023 13:07:18	[pid 2468:tid 348] AH00354: Child: Starting 150 worker threads.	not...	N/A
0.0.0.0	06/06/2023 10:03:37	[pid 3024:tid 320] AH01909: www.example.com:443:O server certificate does NOT include an ID which matches the server name	warn	N/A
0.0.0.0	06/06/2023 10:03:37	[pid 3024:tid 320] AH00098: pid file C:\xampp\apache\logs\httpd.pid overwritten -- Unclean shutdown of previous Apache run?	warn	N/A
0.0.0.0	06/06/2023 10:03:37	[pid 3024:tid 320] AH01909: www.example.com:443:O server certificate does NOT include an ID which matches the server name	warn	N/A
0.0.0.0	06/06/2023 10:03:38	[pid 3024:tid 320] AH00455: Apache/2.4.56 (Win64) OpenSSL/1.1.1t PHP/8.0.28 configured -- resuming normal operations	not...	N/A
0.0.0.0	06/06/2023 10:03:38	[pid 3024:tid 320] AH00456: Apache Lounge VS16 Server built: Mar 7 2023 13:21:03	not...	N/A
0.0.0.0	06/06/2023 10:03:38	[pid 3024:tid 320] AH00094: Command line: 'C:\xampp\apache\bin\httpd.exe -d C:\xampp\apache'	not...	N/A
0.0.0.0	06/06/2023 10:03:38	[pid 3024:tid 320] AH00418: Parent: Created child process 4052	not...	N/A
0.0.0.0	06/06/2023 10:03:38	[pid 4052:tid 336] AH01909: www.example.com:443:O server certificate does NOT include an ID which matches the server name	warn	N/A
0.0.0.0	06/06/2023 10:03:38	[pid 4052:tid 336] AH01909: www.example.com:443:O server certificate does NOT include an ID which matches the server name	warn	N/A
0.0.0.0	06/06/2023 10:03:38	[pid 4052:tid 336] AH00354: Child: Starting 150 worker threads.	not...	N/A
	08/06/2023 23:50:58	PHP Parse error: syntax error, unexpected token ">" in C:\xampp\htdocs\devops\viewstudents.php on line 28, referer: http://loc...	error	N/A

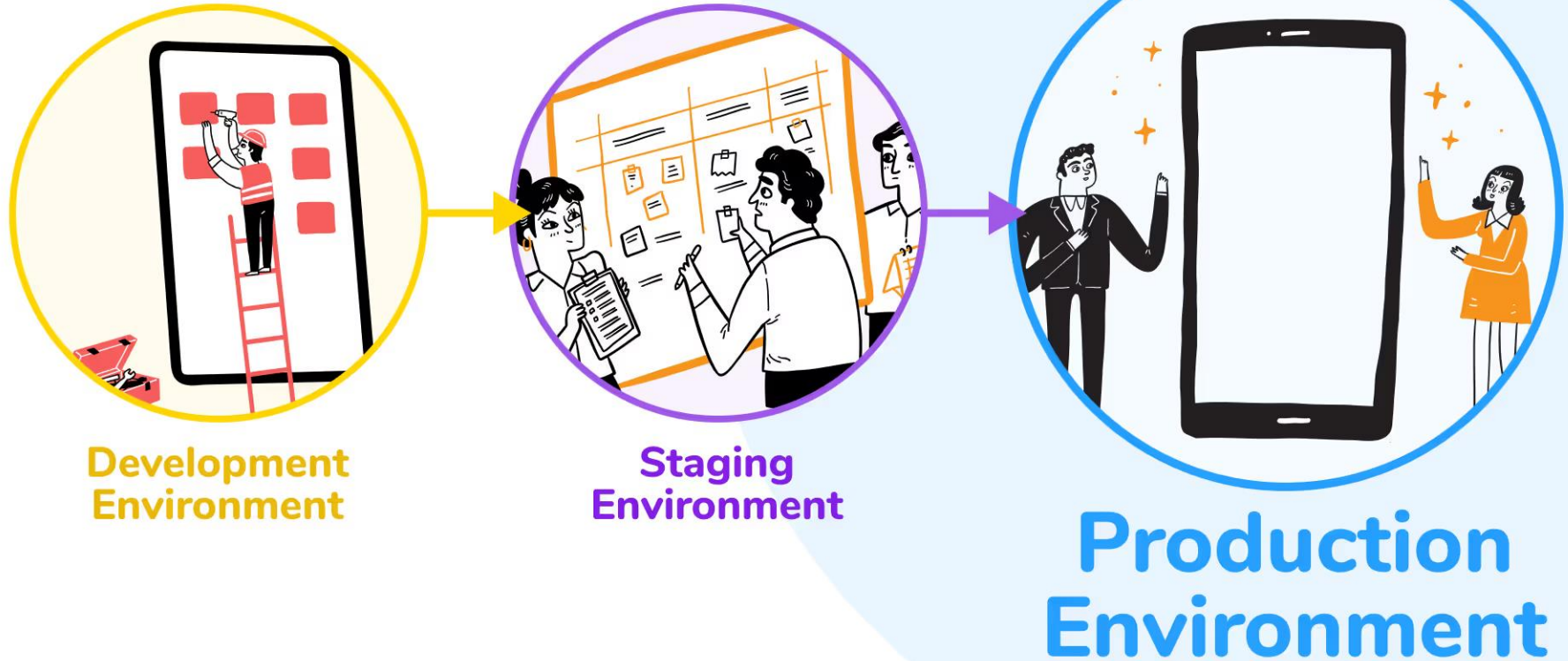
Students diminta untuk melakukan analisis **error log** **Apache**, identifikasi masalah keamanan dan kinerja web server.

Monitoring Security Issues

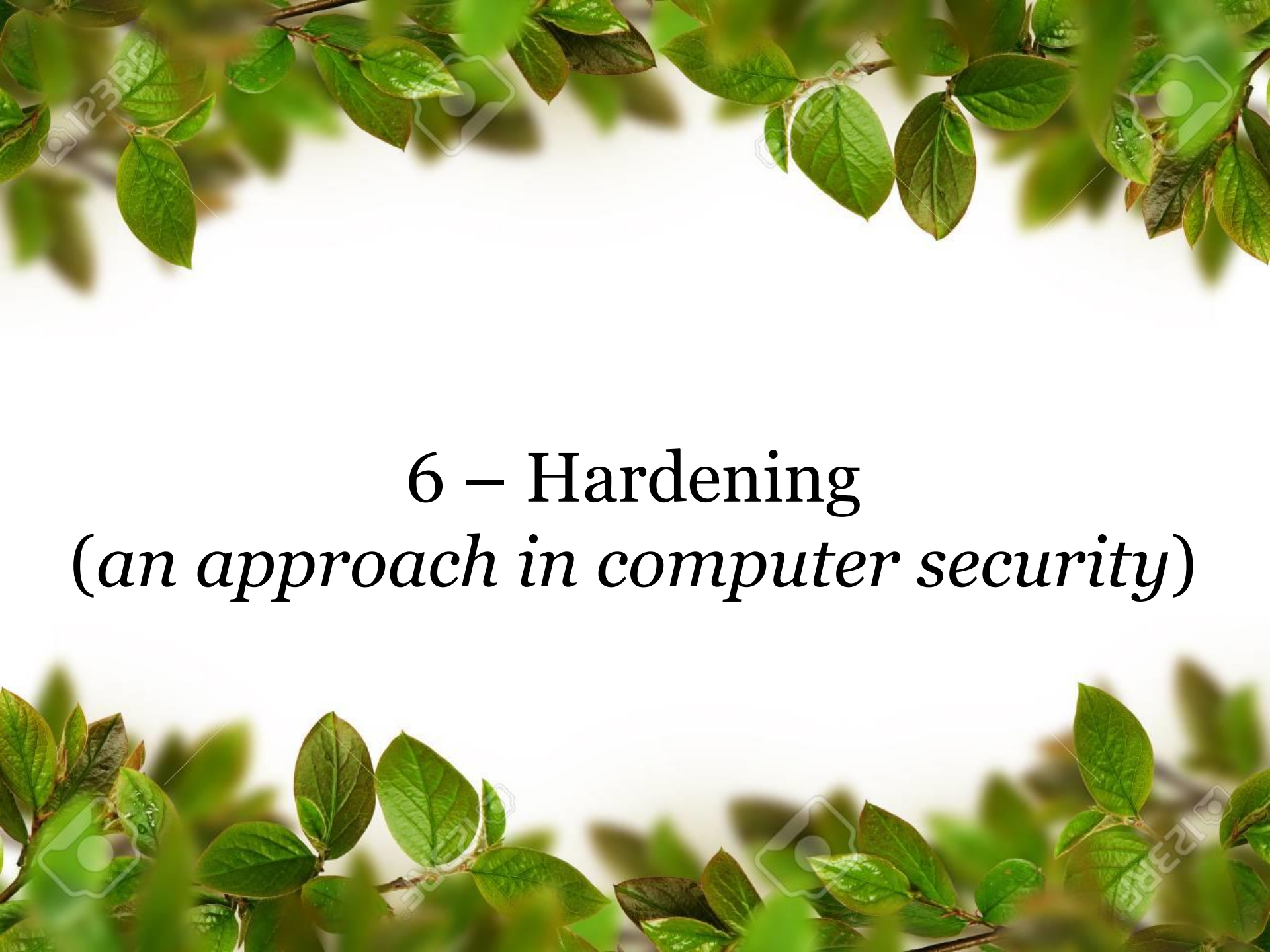
Remember this !!!

Error/auth Log Apache dapat memberikan informasi tentang upaya serangan atau aktivitas mencurigakan. Beberapa serangan atau celah keamanan dapat menghasilkan pesan *error* tertentu. Dengan memantau *log error/auth*, kita dapat mendeteksi aktivitas yang mencurigakan dan mengambil langkah-langkah untuk meningkatkan keamanan *server*.





Menurut anda, apakah terdapat perbedaan pada saat melakukan monitoring error/security issues di dua environment ini?



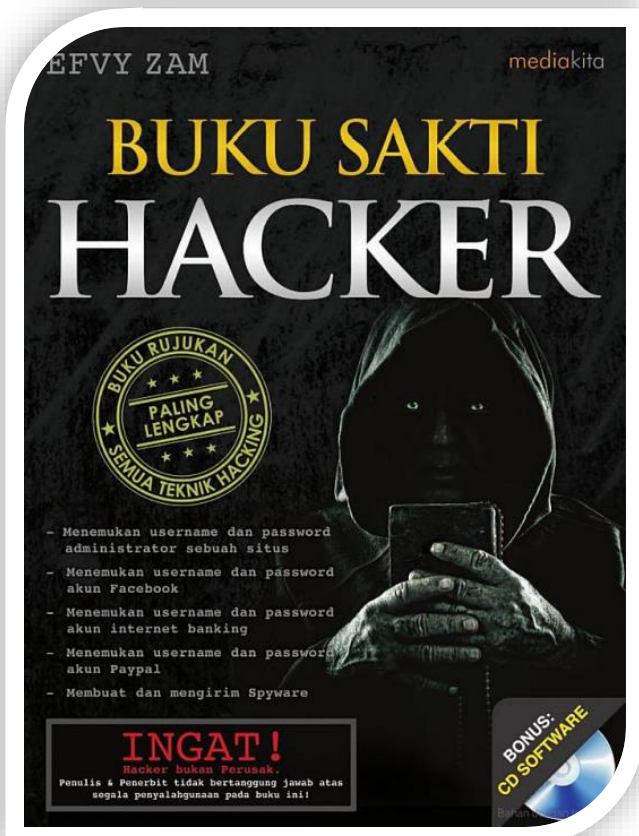
6 – Hardening

(an approach in computer security)

Apakah ada sistem yang benar-benar aman 100%?

Port Scanning

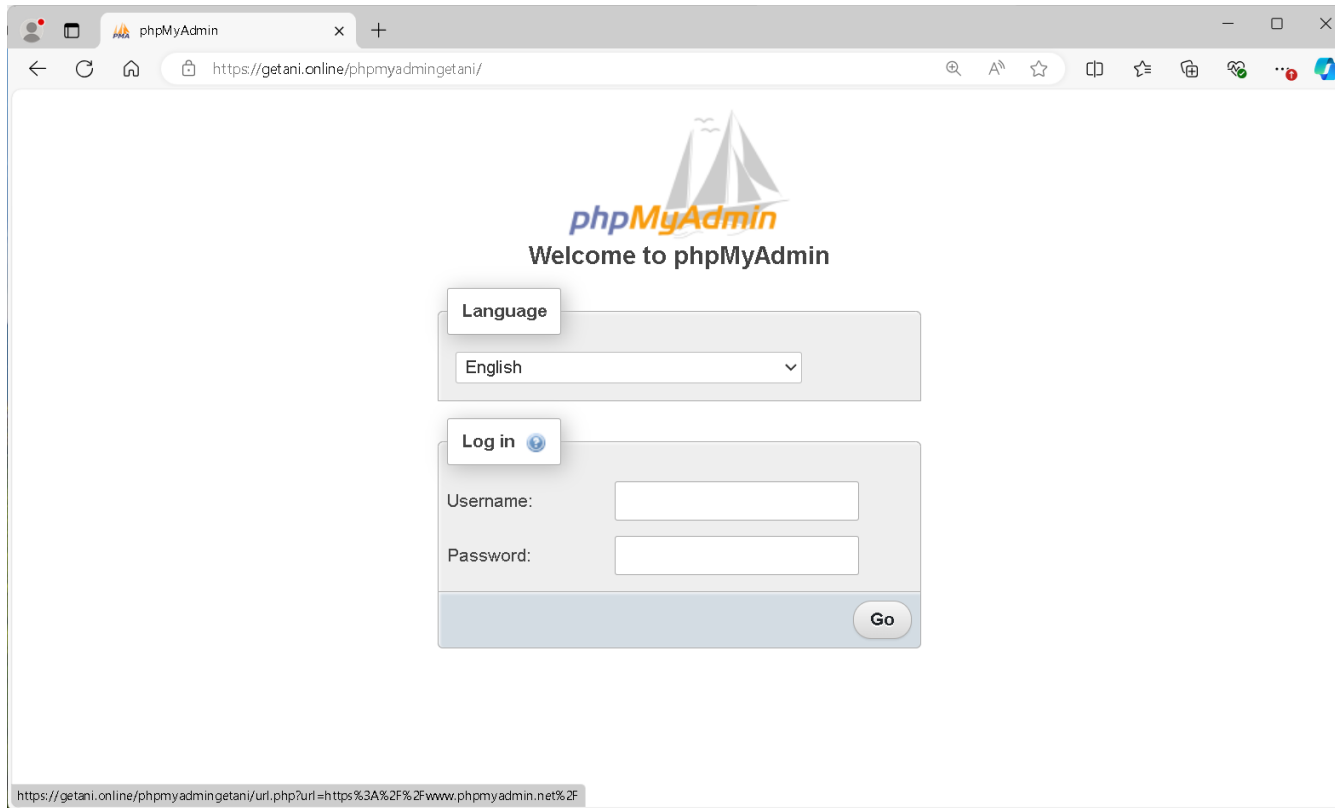
4



Jika diibaratkan sebuah rumah, port adalah pintu dan jendela rumah tempat keluar masuknya data. Secara logika, tidak ada sistem yang aman 100%. Apabila sebuah sistem aman 100%, tentu saja semua pintu dan jendela akan ditutup semua. Ibaratnya, jika pintu internet tidak dibuka, Anda pun tidak akan bisa menghubungkan komputer dengan internet. Jadi, bisa kita katakan bahwa port adalah pintu keluar masuknya paket data. }

Secara garis besar, port dapat dibagi dua bagian. Yang pertama adalah port fisik (*physical port*) yang merupakan port di bagian belakang CPU, seperti port serial, dan port monitor. Lalu ada pula port perangkat lunak (*software port*), merupakan port yang digunakan oleh software untuk melakukan koneksi dengan komputer lain.

Port juga mengidentifikasi sebuah proses tertentu dimana sebuah server dapat memberikan sebuah layanan kepada klien atau bagaimana sebuah klien dapat mengakses sebuah layanan yang ada dalam server. Ada banyak port yang terdapat pada sebuah komputer, apalagi sewaktu terhubung dengan internet. Beberapa port yang umum adalah port 80 (HTTP), yaitu port untuk membuka sebuah halaman website, port 20 (FTP) untuk melakukan upload maupun download file, port 110 (POP3) untuk menerima email. Serta masih banyak jenis port lainnya.



Pada saat kalian buat *cloud server*, phpMyAdmin sudah diinstall dan sudah bisa di akses via **public IP address** nantinya. Apakah kalian berpikir bahwa ***login form*** ini akan aman-aman saja tanpa serangan atau aktivitas mencurigakan???

```
semmy@presensi: ~  
Sep 16 00:07:51 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:07:54 presensi phpMyAdmin[18162]: message repeated 4 times: [ user denied: root (mysql-denied) from 192.3.228.147]  
Sep 16 00:07:55 presensi phpMyAdmin[18162]: user denied: phpmyadmin (mysql-denied) from 192.3.228.147  
Sep 16 00:07:55 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:07:56 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:07:56 presensi phpMyAdmin[18162]: user denied: admin (mysql-denied) from 192.3.228.147  
Sep 16 00:07:57 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:07:57 presensi phpMyAdmin[18162]: user denied: web (mysql-denied) from 192.3.228.147  
Sep 16 00:07:58 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:07:59 presensi phpMyAdmin[18162]: message repeated 3 times: [ user denied: root (mysql-denied) from 192.3.228.147]  
Sep 16 00:08:00 presensi phpMyAdmin[18162]: user denied: wordpress (mysql-denied) from 192.3.228.147  
Sep 16 00:08:00 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:02 presensi phpMyAdmin[18162]: message repeated 3 times: [ user denied: root (mysql-denied) from 192.3.228.147]  
Sep 16 00:08:02 presensi phpMyAdmin[18162]: user denied: developer (mysql-denied) from 192.3.228.147  
Sep 16 00:08:03 presensi phpMyAdmin[18162]: user denied: dbs (mysql-denied) from 192.3.228.147  
Sep 16 00:08:03 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:04 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:04 presensi phpMyAdmin[18162]: user denied: wp (mysql-denied) from 192.3.228.147  
Sep 16 00:08:05 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:09 presensi phpMyAdmin[18162]: message repeated 8 times: [ user denied: root (mysql-denied) from 192.3.228.147]  
Sep 16 00:08:09 presensi phpMyAdmin[18162]: user denied: newuser (mysql-denied) from 192.3.228.147  
Sep 16 00:08:10 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:10 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:11 presensi phpMyAdmin[18162]: user denied: database (mysql-denied) from 192.3.228.147  
Sep 16 00:08:11 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:12 presensi phpMyAdmin[18162]: message repeated 2 times: [ user denied: root (mysql-denied) from 192.3.228.147]  
Sep 16 00:08:13 presensi phpMyAdmin[18162]: user denied: user (mysql-denied) from 192.3.228.147  
Sep 16 00:08:14 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:15 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:16 presensi phpMyAdmin[18162]: user denied: newuser (mysql-denied) from 192.3.228.147  
Sep 16 00:08:16 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:17 presensi phpMyAdmin[18162]: user denied: dev (mysql-denied) from 192.3.228.147  
Sep 16 00:08:17 presensi phpMyAdmin[18162]: user denied: admin (mysql-denied) from 192.3.228.147  
Sep 16 00:08:18 presensi phpMyAdmin[18162]: user denied: admin (mysql-denied) from 192.3.228.147  
Sep 16 00:08:19 presensi phpMyAdmin[18166]: user denied: admin (mysql-denied) from 192.3.228.147  
Sep 16 00:08:20 presensi phpMyAdmin[18166]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:20 presensi phpMyAdmin[18166]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:21 presensi phpMyAdmin[18166]: user denied: phpmyadmin (mysql-denied) from 192.3.228.147  
Sep 16 00:08:22 presensi phpMyAdmin[18166]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:26 presensi phpMyAdmin[18166]: message repeated 7 times: [ user denied: root (mysql-denied) from 192.3.228.147]
```

Hardening is an *approach in computer security* that aims to reduce security risks by reducing potential system vulnerabilities. While no solution can guarantee complete security, hardening is still an important step to protect systems and data.

Hardening hanyalah sebuah pendekatan dalam keamanan komputer yang bertujuan untuk mengurangi risiko keamanan dengan mengurangi potensi kerentanan sistem.

Jelaskan beberapa Langkah-Langkah **Hardening** di server yang anda ketahui untuk mengurangi potential system vulnerabilities?



7 – SQLmap

Alat pengujian penetrasi (*penetration testing*)





Peringatan Etika & Hukum



- 1) Sqlmap adalah alat pengujian penetrasi (*penetration testing*) yang dapat merusak atau melanggar privasi jika digunakan tanpa izin.
- 2) Jangan pernah menggunakan sqlmap terhadap situs/web yang tidak kita miliki atau tanpa izin eksplisit.
- 3) Materi ini hanya memberikan pengenalan tentang kerentanan keamanan. Penerapan pengujian hanya berlaku untuk localhost (**misalnya**: aplikasi web lokal untuk pengembangan /pengujian) ataupun situs milik pribadi.
- 4) Menggunakan Sqlmap pada situs web yang tidak kita miliki melanggar hukum (UU ITE di Indonesia, Computer Fraud and Abuse Act di US, dll).

- 
- 1) **Penetration testing** (*pentest*) is an **authorized simulation of real-world attacker techniques**, used to hunt for and highlight vulnerabilities in our networks, applications and devices.
- 2) This is done by testing in a **controlled environment** without compromising routine business activities.
- 3) There are many forms of Penetration Testing services, including :
- (a) External Penetration Testing,
 - (b) Internal Penetration Testing,
 - (c) Web Application Security Testing,
 - (d) Mobile Application Security Assessment,
 - (e) Wireless Network Security Assessment and
 - (f) Cloud Penetration Testing.



SOFTWARE TESTING

Pentest atau simulasi serangan cyber dapat dilakukan secara sah dan terkendali terhadap sistem, jaringan, aplikasi, atau infrastruktur TI untuk mengidentifikasi dan mengeksploitasi kerentanan keamanan sebelum attacker (*pihak penyerang*) melakukan serangan.

Install Sqlmap di Windows

1: Instal Python

2: Download/clone sqlmap

git clone https://github.com/sqlmapproject/sqlmap.git

The screenshot shows the GitHub repository page for sqlmapproject/sqlmap. The repository is public and has 36.4k stars and 6.2k forks. The repository is maintained by stamparm and has 10,461 commits. The repository is described as an "Automatic SQL injection and database takeover tool". The repository includes a README, a license, a code of conduct, and contributing guidelines. The repository is tagged with various keywords including python, database, detection, sql-injection, pentesting, exploitation, sqlmap, takeover, and vulnerability-scanner. The repository is also tagged with 1.1k watching.

GitHub - sqlmapproject/sqlmap

github.com/sqlmapproject/sqlmap?spm=a2ty_o01.29997173.0.0.75b25171wqBJ3W

Platform Solutions Resources Open Source Enterprise Pricing

sqlmapproject / sqlmap Public

Sponsor Notifications Fork 6.2k Star 36.4k

Code Issues 39 Pull requests 15 Actions Projects Wiki Security Insights

master 2 Branches 133 Tags

Go to file

Code

About

Automatic SQL injection and database takeover tool

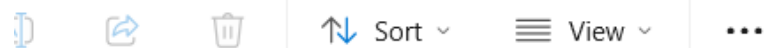
sqlmap.org

python database detection sql-injection pentesting exploitation sqlmap takeover vulnerability-scanner

Readme View license Code of conduct Contributing Activity Custom properties 36.4k stars 1.1k watching

File	Commit Message	Time Ago
.github	Bump python-version for GitHub tests	11 months ago
data	Minor update	3 days ago
doc	Removing some dead code	5 days ago
extra	Minor update of vuln testing	2 weeks ago
lib	Minor update	3 days ago
plugins	Minor patch for Snowflake implementation (#5980)	last week
tamper	Year bump	3 weeks ago
thirdparty	Removing some dead code	5 days ago
.gitattributes	Renaming Twitter to X	2 years ago

This PC > Acer (C:) > Users > USER > sqlmap >



Name	Date modified	Type	Size
.git	16/09/2025 13:17	File folder	
.github	16/09/2025 13:17	File folder	
data	16/09/2025 13:17	File folder	
doc	16/09/2025 13:17	File folder	
extra	16/09/2025 13:17	File folder	
lib	16/09/2025 13:17	File folder	
plugins	16/09/2025 13:17	File folder	
tamper	16/09/2025 13:17	File folder	
thirdparty	16/09/2025 13:17	File folder	
.gitattributes	16/09/2025 13:17	Git Attributes Sour...	1 KB
.gitignore	16/09/2025 13:17	Git Ignore Source ...	1 KB
LICENSE	16/09/2025 13:17	File	19 KB
README.md	16/09/2025 13:17	Markdown Source ...	6 KB
sqlmap.conf	16/09/2025 13:17	CONF File	23 KB
sqlmap.py	16/09/2025 13:17	PY File	26 KB
sqlmapapi.py	16/09/2025 13:17	PY File	5 KB
sqlmapapi.yaml	16/09/2025 13:17	Yaml Source File	7 KB

tests.yml passing python 2.7|3.x license GPLv2 x @sqlmap

sqlmap is an open source penetration testing tool that automates the process of detecting and exploiting SQL injection flaws and taking over of database servers. It comes with a powerful detection engine, many niche features for the ultimate penetration tester, and a broad range of switches including database fingerprinting, over data fetching from the database, accessing the underlying file system, and executing commands on the operating system via out-of-band connections.

Screenshots

```
$ python sqlmap.py -u "http://172.16.112.128/sqlmap/mysql/get_int.php?id=1" --batch
```



{1.3.4.44#dev}
<http://sqlmap.org>

```
[!] legal disclaimer: Usage of sqlmap for attacking targets without prior mutual consent is illegal. It is the end user's responsibility to obey all applicable local, state and federal laws. Developers assume no liability and are not responsible for any misuse or damage caused by this program
```

```
[*] starting @ 10:34:28 /2019-04-30/
```

```
[10:34:28] [INFO] testing connection to the target URL
[10:34:28] [INFO] heuristics detected web page charset 'ascii'
[10:34:28] [INFO] checking if the target is protected by some kind of WAF/IPS
[10:34:28] [INFO] testing if the target URL content is stable
[10:34:29] [INFO] target URL content is stable
[10:34:29] [INFO] testing if GET parameter 'id' is dynamic
[10:34:29] [INFO] GET parameter 'id' appears to be dynamic
[10:34:29] [INFO] heuristic (basic) test shows that GET parameter 'id' might be injectable (possible DBMS: 'MySQL')
[10:34:29] [INFO] heuristic (XSS) test shows that GET parameter 'id' might be vulnerable to cross-site scripting (XSS) attacks
```

Server Firewall

UFW (*Uncomplicated Firewall*) Adalah firewall berbasis antarmuka sederhana untuk iptables di sistem operasi Linux, terutama yang berbasis Debian/Ubuntu.

To	Action	From
--	-----	----
22/tcp	DENY	Anywhere
80/tcp	ALLOW	Anywhere
443/tcp	ALLOW	Anywhere
3306/tcp	DENY	Anywhere
8899/tcp	DENY	Anywhere
Anywhere	DENY	196.251.83.136
8899/tcp	DENY	180.251.192.195
Anywhere	DENY	52.169.188.222
22/tcp (v6)	DENY	Anywhere (v6)
80/tcp (v6)	ALLOW	Anywhere (v6)
443/tcp (v6)	ALLOW	Anywhere (v6)
3306/tcp (v6)	DENY	Anywhere (v6)
8899/tcp (v6)	DENY	Anywhere (v6)

Tujuan utama *penetration testing* yaitu untuk menemukan celah keamanan (*vulnerabilities*) seperti SQL injection, XSS, konfigurasi salah, dll. Digunakan juga untuk mengevaluasi efektivitas kontrol keamanan yang sudah ada (*firewall, autentikasi, enkripsi, dll*).

Jelaskan menurut anda, bagaimana cara mengidentifikasi dan menangani SQLmap?



Students diminta untuk mengikuti *step by step* yang telah dibuat, selanjutnya membuat report dalam format PDF dan submit di Google Classroom !!!

END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

